



Massachusetts Institute of Technology

MIT  $\$(\hat{w})\$$

Benjamin Qi, Spencer Compton, Zhezheng Luo

adapted from KACTL and MIT NULL

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- 2 Mathematics
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Contest (1)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                  |
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| TemplateShort.cpp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 77aa31, 44 lines |
| <pre>#include &lt;bits/stdc++.h&gt; using namespace std;  using ll = long long; using db = long double; // or double if tight TL using str = string;  using pi = pair&lt;int,int&gt;; #define mp make_pair #define f first #define s second  #define tcT template&lt;class T tcT&gt; using V = vector&lt;T&gt;; tcT, size_t SZ&gt; using AR = array&lt;T,SZ&gt;; using vi = V&lt;int&gt;; using vb = V&lt;bool&gt;; using vpi = V&lt;pi&gt;;  #define sz(x) int((x).size()) #define all(x) begin(x), end(x) #define sor(x) sort(all(x)) #define rsz resize #define pb push_back #define ft front() #define bk back()  #define FOR(i,a,b) for (int i = (a); i &lt; (b); ++i) #define F0R(i,a) FOR(i,0,a) #define ROF(i,a,b) for (int i = (b)-1; i &gt;= (a); --i) #define R0F(i,a) ROF(i,0,a) #define rep(a) F0R(_,a) #define each(a,x) for (auto&amp; a: x)  const int MOD = 1e9+7; const db PI = acos((db)-1); mt19937 rng(0); // or mt19937_64</pre> |                  |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | tcT> <b>bool</b> ckmin(T& a, <b>const</b> T& b) {<br><b>return</b> b < a ? a = b, 1 : 0; } // set a = min(a,b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 1  | tcT> <b>bool</b> ckmax(T& a, <b>const</b> T& b) {<br><b>return</b> a < b ? a = b, 1 : 0; } // set a = max(a,b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 2  | <b>int</b> main() { cin.tie(0)->sync_with_stdio(0); }                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 3  | <div><div>.bashrc</div><div>3 lines</div></div> <pre><b>alias</b> clr="printf '\33c'" co() { g++ -std=c++17 -O2 -Wall -Wextra -Wshadow -Wconversion -     ↪o \$1 \$1.cpp; } run() { co \$1 &amp;&amp; ./\$1; }</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 6  | <div><div>hash.sh</div><div>3 lines</div></div> <pre># Hash file ignoring whitespace and comments. Verifies that # code was correctly typed. Usage: 'sh hash.sh &lt; A.cpp' cpp -dD -P -fpreprocessed tr -d '[:space:]' md5sum cut -c-6</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 8  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 14 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 19 | <div><div>stress.sh</div><div>10 lines</div></div> <pre># A and B are executables you want to compare, gen takes int # as command line arg. Usage: 'sh stress.sh' <b>for</b>((i = 1; ; ++i)); <b>do</b>     <b>echo</b> \$i     ./gen \$i &gt; int     ./A &lt; int &gt; out1     ./B &lt; int &gt; out2     diff -w out1 out2    <b>break</b>     # diff -w &lt;(.A &lt; int) &lt;(.B &lt; int)    <b>break</b> <b>done</b></pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 21 | <div><div>troubleshoot.txt</div><div>75 lines</div></div> <p>General:<br/>Write down most of your thoughts, even if you’re not sure whether they’re useful.<br/>Give your variables (and files) meaningful names.<br/>Stay organized and don’t leave papers all over the place!<br/>You should know what your code is doing ...</p> <p>Pre-submit:<br/>Write a few simple test cases if sample is not enough.<br/>Are time limits close? If so, generate max cases.<br/>Is the memory usage fine?<br/>Could anything overflow?<br/>Remove debug output.<br/>Make sure to submit the right file.</p> <p>Wrong answer:<br/>Print your solution! Print debug output as well.<br/>Read the full problem statement again.<br/>Have you understood the problem correctly?<br/>Are you sure your algorithm works?<br/>Try writing a slow (but correct) solution.<br/>Can your algorithm handle the whole range of input?<br/>Did you consider corner cases (ex. n=1)?<br/>Is your output format correct? (including whitespace)<br/>Are you clearing all data structures between test cases?<br/>Any uninitialized variables?<br/>Any undefined behavior (array out of bounds)?<br/>Any overflows or NaNs (or shifting ll by &gt;=64 bits)?<br/>Confusing N and M, i and j, etc.?<br/>Confusing ++i and i++?<br/>Return vs continue vs break?<br/>Are you sure the STL functions you use work as you think?<br/>Add some assertions, maybe resubmit.<br/>Create some test cases to run your algorithm on.</p> |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <p>Go through the algorithm for a simple case.<br/>Go through this list again.<br/>Explain your algorithm to a teammate.<br/>Ask the teammate to look at your code.<br/>Go for a small walk, e.g. to the toilet.<br/>Rewrite your solution from the start or let a teammate do it.</p> <p>Geometry:<br/>Work with ints if possible.<br/>Correctly account for numbers close to (but not) zero. Related:<br/>for functions like acos make sure absolute val of input is not (slightly) greater than one.<br/>Correctly deal with vertices that are collinear, concyclic, coplanar (in 3D), etc.<br/>Subtracting a point from every other (but not itself)?</p> <p>Runtime error:<br/>Have you tested all corner cases locally?<br/>Any uninitialized variables?<br/>Are you reading or writing outside the range of any vector?<br/>Any assertions that might fail?<br/>Any possible division by 0? (mod 0 for example)<br/>Any possible infinite recursion?<br/>Invalidated pointers or iterators?<br/>Are you using too much memory?<br/>Debug with resubmits (e.g. remapped signals, see Various).</p> <p>Time limit exceeded:<br/>Do you have any possible infinite loops?<br/>What’s your complexity? Large TL does not mean that something simple (like NlogN) isn’t intended.<br/>Are you copying a lot of unnecessary data? (References)<br/>Avoid vector, map. (use arrays/unordered_map)<br/>How big is the input and output? (consider FastIO)<br/>What do your teammates think about your algorithm?<br/>Calling count() on multiset?</p> <p>Memory limit exceeded:<br/>What is the max amount of memory your algorithm should need?<br/>Are you clearing all data structures between test cases?<br/>If using pointers try BumpAllocator.</p> <h2>Mathematics (2)</h2> <h3>2.1 Trigonometry</h3> <div><div><math display="block">\sin(v+w) = \sin v \cos w + \cos v \sin w</math><math display="block">\cos(v+w) = \cos v \cos w - \sin v \sin w</math><math display="block">\tan(v+w) = \frac{\tan v + \tan w}{1 - \tan v \tan w}</math><math display="block">\sin v + \sin w = 2 \sin \frac{v+w}{2} \cos \frac{v-w}{2}</math><math display="block">\cos v + \cos w = 2 \cos \frac{v+w}{2} \cos \frac{v-w}{2}</math></div><div><math display="block">a \cos x + b \sin x = r \cos(x - \phi)</math><math display="block">a \sin x + b \cos x = r \sin(x + \phi)</math></div></div> <p>where <math>r = \sqrt{a^2 + b^2}, \phi = \operatorname{atan2}(b, a)</math>.</p> |
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## 2.2 Geometry

### 2.2.1 Triangles

Side lengths:  $a, b, c$

Semiperimeter:  $s = \frac{a+b+c}{2}$

Area:  $A = \sqrt{s(s-a)(s-b)(s-c)}$

Circumradius:  $R = \frac{abc}{4A}$

Inradius:  $r = \frac{A}{p}$

Length of median (divides triangle into two equal-area triangles):

$$m_a = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$$

Length of bisector (divides angles in two):

$$s_a = \sqrt{bc \left[ 1 - \left( \frac{a}{b+c} \right)^2 \right]}$$

Law of sines:  $\frac{\sin \alpha}{a} = \frac{\sin \beta}{b} = \frac{\sin \gamma}{c} = \frac{1}{2R}$

Law of cosines:  $a^2 = b^2 + c^2 - 2bc \cos \alpha$

Law of tangents:  $\frac{a+b}{a-b} = \frac{\tan \frac{\alpha+\beta}{2}}{\tan \frac{\alpha-\beta}{2}}$

## 2.3 Derivatives/Integrals

$$\begin{aligned} \frac{d}{dx} \arcsin x &= \frac{1}{\sqrt{1-x^2}} & \frac{d}{dx} \arccos x &= -\frac{1}{\sqrt{1-x^2}} \\ \frac{d}{dx} \tan x &= 1 + \tan^2 x & \frac{d}{dx} \arctan x &= \frac{1}{1+x^2} \\ \int \tan ax &= -\frac{\ln|\cos ax|}{a} & \int x \sin ax &= \frac{\sin ax - ax \cos ax}{a^2} \\ \int e^{-x^2} &= \frac{\sqrt{\pi}}{2} \operatorname{erf}(x) & \int x e^{ax} dx &= \frac{e^{ax}}{a^2} (ax - 1) \end{aligned}$$

Integration by parts:

$$\int_a^b f(x)g(x)dx = [F(x)g(x)]_a^b - \int_a^b F(x)g'(x)dx$$

## 2.4 Sums/Series

$$\begin{aligned} \ln(1+x) &= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots, \quad (-1 < x \leq 1) \\ \sqrt{1+x} &= 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{2x^3}{32} - \frac{5x^4}{128} + \dots, \quad (-1 \leq x \leq 1) \\ \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots, \quad (-\infty < x < \infty) \\ \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots, \quad (-\infty < x < \infty) \end{aligned}$$

## Data Structures (3)

### 3.1 STL

MapComparator.h

**Description:** example of function object (functor) for map or set

**Usage:** set<int, cmp> s; map<int, int, cmp> m; 5bfa6c, 1 lines

**struct** cmp{**bool** operator() (int l, int r) const { **return** l>r; } };

HashMap.h

**Description:** Hash map with similar API as unordered\_map. Initial capacity must be a power of 2 if provided.

**Usage:** ht<int, int> h {{}, {}, {}, {}, {1<<16}};

**Memory:** ~1.5x unordered map

**Time:** ~3x faster than unordered map

<ext/pb\_ds/assoc.container.hpp> 5872b2, 9 lines

```
using namespace __gnu_pbds;
struct chash {
    const uint64_t C = 1l(4e18*acos(0))+71; // large odd number
    const int RANDOM = rng();
    ll operator() (ll x) const { return __builtin_bswap64((x^
        ↪RANDOM)*C); }
};
template<class K, class V> using ht = gp_hash_table<K, V, chash>;
template<class K, class V> V get(ht<K, V>& u, K x) {
    auto it = u.find(x); return it == end(u) ? 0 : it->s; }
```

OrderStatisticTree.h

**Description:** A set (not multiset!) with support for finding the  $n$ 'th element, and finding the index of an element. Change null\_type to get a map.

**Time:**  $\mathcal{O}(\log N)$

<ext/pb\_ds/assoc.container.hpp> fe2351, 12 lines

```
using namespace __gnu_pbds;
tcT> using Tree = tree<T, null_type, less<T>,
    rb_tree_tag, tree_order_statistics_node_update>;
#define ook order_of_key
#define fbo find_by_order
```

```
void treeExample() {
    Tree<int> t, t2; t.insert(8);
    auto it = t.insert(10).f; assert(it == t.lb(9));
    assert(t.ook(10) == 1 && t.ook(11) == 2 && *t.fbo(0) == 8);
    t.join(t2); // assuming T < T2 or T > T2, merge t2 into t
}
```

LineContainer.h

**Description:** Add lines of the form  $ax + b$ , query maximum  $y$ -coordinate for any  $x$ .

**Time:**  $\mathcal{O}(\log N)$

using T = ll; const T INF = LLONG\_MAX; // a/b rounded down
// ll fdiv(ll a, ll b) { return a/b-((a^b)<0&&a%b); }

```
bool _Q = 0;
struct Line {
    T a, b; mutable T lst;
    T eval(T x) const { return a*x+b; }
    bool operator<(const Line&o) const { return _Q?lst<o.lst:a<o.a; }
    T last_gre(const Line&o) const { assert(a <= o.a);
        // greatest x s.t. a*x+b >= o.a*x+o.b
        return lst=(a==o.a?(b>=o.b?INF:-INF):fdiv(b-o.b,o.a-a)); }
};
```

```
struct LineContainer: multiset<Line> {
    bool isect(iterator it) { auto n_it = next(it);
        if (n_it == end()) return it->lst = INF, 0;
        return it->last_gre(*n_it) >= n_it->lst; }
```

```
void add(T a, T b) {
    auto it = ins({a,b,0}); while (isect(it)) erase(next(it));
    if (it == begin()) return;
    if (isect(--it)) erase(next(it)), isect(it);
    while (it != begin()) {
        --it; if (it->lst < next(it)->lst) break;
        erase(next(it)); isect(it); }
}
T qmax(T x) { assert(!empty());
    _Q = 1; T res = lb({0,0,x})->eval(x); _Q = 0;
    return res; }
};
```

### 3.2 1D Range Queries

RMQ.h

**Description:** 1D range minimum query. If TL is an issue, use arrays instead of vectors and store values instead of indices.

**Memory:**  $\mathcal{O}(N \log N)$

**Time:**  $\mathcal{O}(1)$

a3f881, 19 lines

```
tcT> struct RMQ { // floor(log_2(x))
    int level(int x) { return 31-__builtin_clz(x); }
    V<T> v; V<vi> jmp;
    int cmb(int a, int b) {
        return v[a]==v[b]?min(a,b):(v[a]<v[b]?a:b); }
    void init(const V<T>& _v) {
        v = _v; jmp = {vi(sz(v))};
        iota(all(jmp[0]),0);
        for (int j = 1; 1<<j <= sz(v); ++j) {
            jmp.pb(vi(sz(v)-(1<<j)+1));
            F0R(i, sz(jmp[j])) jmp[j][i] = cmb(jmp[j-1][i],
                jmp[j-1][i+(1<<(j-1))]);
        }
    }
    int index(int l, int r) {
        assert(l <= r); int d = level(r-l+1);
        return cmb(jmp[d][l], jmp[d][r-(1<<d)+1]); }
    T query(int l, int r) { return v[index(l,r)]; }
};
```

SegmentTree.h

**Description:** 1D point update and range query where cmb is any associative operation. seg[1]==query(0,N-1).

**Time:**  $\mathcal{O}(\log N)$

1630f3, 18 lines

```
tcT> struct SegTree { // cmb(ID,b) = b
    const T ID{}; T cmb(T a, T b) { return a+b; }
    int n; V<T> seg;
    void init(int _n) { // upd, query also work if n = _n
        for (n = 1; n < _n; ) n *= 2;
        seg.assign(2*n, ID); }
    void pull(int p) { seg[p] = cmb(seg[2*p], seg[2*p+1]); }
    void upd(int p, T val) { // set val at position p
        seg[p += n] = val; for (p /= 2; p; p /= 2) pull(p); }
    T query(int l, int r) { // zero-indexed, inclusive
        T ra = ID, rb = ID;
        for (l += n, r += n+1; l < r; l /= 2, r /= 2) {
            if (l&1) ra = cmb(ra, seg[l++]);
            if (r&1) rb = cmb(seg[--r], rb);
        }
        return cmb(ra, rb);
    }
};
```

LazySegmentTree.h

**Description:** 1D range increment and sum query.

**Time:**  $\mathcal{O}(\log N)$

78a06d, 26 lines

```
tcT, int SZ> struct LazySeg {
```

```
static_assert(pct(SZ) == 1); // SZ must be power of 2
const T ID{}; T cmb(T a, T b) { return a+b; }
T seg[2*SZ], lazy[2*SZ];
LazySeg() { F0R(i,2*SZ) seg[i] = lazy[i] = ID; }
void push(int ind, int L, int R) {
    seg[ind] += (R-L+1)*lazy[ind]; // dependent on operation
    if (L != R) F0R(i,2) lazy[2*ind+i] += lazy[ind];
    lazy[ind] = 0;
} // recalc values for current node
void pull(int ind){seg[ind]=cmb(seg[2*ind],seg[2*ind+1]);}
void build() { ROF(i,1,SZ) pull(i); }
void upd(int lo,int hi,T inc,int ind=1,int L=0, int R=SZ-1) {
    push(ind,L,R); if (hi < L || R < lo) return;
    if (lo <= L && R <= hi) {
        lazy[ind] = inc; push(ind,L,R); return; }
    int M = (L+R)/2; upd(lo,hi,inc,2*ind,L,M);
    upd(lo,hi,inc,2*ind+1,M+1,R); pull(ind);
}
T query(int lo, int hi, int ind=1, int L=0, int R=SZ-1) {
    push(ind,L,R); if (lo > R || L > hi) return ID;
    if (lo <= L && R <= hi) return seg[ind];
    int M = (L+R)/2; return cmb(query(lo,hi,2*ind,L,M),
        query(lo,hi,2*ind+1,M+1,R));
}
};
```

**PSeg.h**  
**Description:** Persistent min segtree with lazy updates, no propagation. If making d a vector then save the results of upd and build in local variables first to avoid issues when vector resizes in C++14 or lower.  
**Memory:**  $\mathcal{O}(N+Q\log N)$

```
tcT, int SZ> struct pseg {
    static const int LIM = 2e7;
    struct node {
        int l, r; T val = 0, lazy = 0;
        void inc(T x) { lazy += x; }
        T get() { return val+lazy; }
    };
    node d[LIM]; int nex = 0;
    int copy(int c) { d[nex] = d[c]; return nex++; }
    T cmb(T a, T b) { return min(a,b); }
    void pull(int c) { d[c].val =
        cmb(d[d[c].l].get(), d[d[c].r].get()); }
    /// MAIN FUNCTIONS
    T query(int c, int lo, int hi, int L, int R) {
        if (lo <= L && R <= hi) return d[c].get();
        if (R < lo || hi < L) return MOD;
        int M = (L+R)/2;
        return d[c].lazy+cmb(query(d[c].l,lo,hi,L,M),
            query(d[c].r,lo,hi,M+1,R));
    }
    int upd(int c, int lo, int hi, T v, int L, int R) {
        if (R < lo || hi < L) return c;
        int x = copy(c);
        if (lo <= L && R <= hi) { d[x].inc(v); return x; }
        int M = (L+R)/2;
        d[x].l = upd(d[x].l,lo,hi,v,L,M);
        d[x].r = upd(d[x].r,lo,hi,v,M+1,R);
        pull(x); return x;
    }
    int build(const V<T>& arr, int L, int R) {
        int c = nex++;
        if (L == R) {
            if (L < sz(arr)) d[c].val = arr[L];
            return c;
        }
        int M = (L+R)/2;
        d[c].l = build(arr,L,M), d[c].r = build(arr,M+1,R);
```

```
pull(c); return c;
}
vi loc; /// PUBLIC
void upd(int lo, int hi, T v) {
    loc.pb(upd(loc.bk,lo,hi,v,0,SZ-1)); }
T query(int ti, int lo, int hi) {
    return query(loc[ti],lo,hi,0,SZ-1); }
void build(const V<T>&arr) {loc.pb(build(arr,0,SZ-1)); }
};

Treap.h  

Description: Easy BBST. Use split and merge to implement insert and delete.  

Time:  $\mathcal{O}(\log N)$ 
```

```
using pt = struct tnode*;
struct tnode {
    int pri, val; pt c[2]; // essential
    int sz; ll sum; // for range queries
    bool flip = 0; // lazy update
    tnode(int _val) {
        pri = rng(); sum = val = _val;
        sz = 1; c[0] = c[1] = nullptr;
    }
    ~tnode() { F0R(i,2) delete c[i]; }
};
int getsz(pt x) { return x?x->sz:0; }
ll getsum(pt x) { return x?x->sum:0; }
pt prop(pt x) { // lazy propagation
    if (!x || !x->flip) return x;
    swap(x->c[0],x->c[1]);
    x->flip = 0; F0R(i,2) if (x->c[i]) x->c[i]->flip ^= 1;
    return x;
}
pt calc(pt x) {
    pt a = x->c[0], b = x->c[1];
    assert(!x->flip); prop(a), prop(b);
    x->sz = 1+getsz(a)+getsz(b);
    x->sum = x->val+getsum(a)+getsum(b);
    return x;
}
void tour(pt x, vi& v) { // print values of nodes,
    if (!x) return; // inorder traversal
    prop(x); tour(x->c[0],v); v.pb(x->val); tour(x->c[1],v);
}
pair<pt,pt> split(pt t, int v) { // >= v goes to the right
    if (!t) return {t,t};
    prop(t);
    if (t->val >= v) {
        auto p = split(t->c[0], v); t->c[0] = p.s;
        return {p.f,calc(t)};
    } else {
        auto p = split(t->c[1], v); t->c[1] = p.f;
        return {calc(t),p.s};
    }
}
pair<pt,pt> splitsz(pt t, int sz) { // sz nodes go to left
    if (!t) return {t,t};
    prop(t);
    if (getsz(t->c[0]) >= sz) {
        auto p = splitsz(t->c[0],sz); t->c[0] = p.s;
        return {p.f,calc(t)};
    } else {
        auto p=splitsz(t->c[1],sz-getsz(t->c[0])-1); t->c[1]=p.f;
        return {calc(t),p.s};
    }
}
pt merge(pt l, pt r) { // keys in l < keys in r
    if (!l || !r) return l?r;
```

```
prop(l), prop(r); pt t;
if (l->pri > r->pri) l->c[1] = merge(l->c[1],r), t = l;
else r->c[0] = merge(l,r->c[0]), t = r;
return calc(t);
}
pt ins(pt x, int v) { // insert v
    auto a = split(x,v), b = split(a.s,v+1);
    return merge(a.f,merge(new tnode(v),b.s)); }
pt del(pt x, int v) { // delete v
    auto a = split(x,v), b = split(a.s,v+1);
    return merge(a.f,b.s); }
```

## Number Theory (4)

### 4.1 Modular Arithmetic

```
ModIntShort.h  

Description: Modular arithmetic. Assumes MOD is prime.  

Usage: mi a = MOD+5; inv(a); // 400000003
2672f9, 24 lines

template<int MOD, int RT> struct mint {
    static const int mod = MOD;
    static constexpr mint rt() { return RT; } // primitive root
    int v;
    explicit operator int() const { return v; }
    mint():v(0) {}
    mint(ll _v):v(int(_v%MOD)) { v += (v<0)*MOD; }
    mint& operator+=(mint o) {
        if ((v += o.v) >= MOD) v -= MOD;
        return *this; }
    mint& operator-=(mint o) {
        if ((v -= o.v) < 0) v += MOD;
        return *this; }
    mint& operator*=(mint o) {
        v = int((ll)v*o.v%MOD); return *this; }
    friend mint pow(mint a, ll p) { assert(p >= 0);
        return p==0?1:pow(a*a,p/2)*(p&1?a:1); }
    friend mint inv(mint a) { assert(a.v != 0); return pow(a,MOD
        ↷-2); }
    friend mint operator+(mint a, mint b) { return a += b; }
    friend mint operator-(mint a, mint b) { return a -= b; }
    friend mint operator*(mint a, mint b) { return a *= b; }
};
using mi = mint<(int)1e9+7, 5>;
using vmi = V<mi>;
```

**ModFact.h**  
**Description:** Combinations modulo a prime MOD. Assumes  $2 \leq N \leq MOD$ .  
**Usage:** F.init(10); F.C(6, 4); // 15  
**Time:**  $\mathcal{O}(N)$

```
"ModInt.h"
364271, 13 lines

struct {
    vmi invs, fac, ifac;
    void init(int N) { // idempotent
        invs.rsz(N), fac.rsz(N), ifac.rsz(N);
        invs[1] = fac[0] = ifac[0] = 1;
        FOR(i,2,N) invs[i] = mi(-1ll)MOD/i*(int)invs[MOD%i];
        FOR(i,1,N) fac[i] = fac[i-1]*i, ifac[i] = ifac[i-1]*invs[i
            ↷-1];
        }
    mi C(int a, int b) {
        if (a < b || b < 0) return 0;
        return fac[a]*ifac[b]*ifac[a-b];
    }
} F;
```

ModMulLL.h  
**Description:** Multiply two 64-bit integers mod another if 128-bit is not available. modMul is equivalent to (ul) ( \_\_int128(a)\*b%mod). Works for  $0 \leq a, b < mod < 2^{63}$ . NOTE : Only works if system supports 80-bit floating point.

| d22eee, 12 lines                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>using ul = uint64_t; // ul modMul(ul a, ul b, const ul mod) { //   ll ret = a*b-mod*(ul)((db)a*b/mod); //   return ret+((ret&lt;0)-(ret&gt;=(ll)mod))*mod; } ul modMul(ul a, ul b, const ul mod){     return __int128(a) * __int128(b) % mod; } ul modPow(ul a, ul b, const ul mod) {     if (b == 0) return 1;     ul res = modPow(a,b/2,mod); res = modMul(res,res,mod);     return b&amp;1 ? modMul(res,a,mod) : res; }</pre> |

FastMod.h  
**Description:** Barrett reduction computes  $a\%b$  about 4 times faster than usual where  $b > 1$  is constant but not known at compile time. Division by  $b$  is replaced by multiplication by  $m$  and shifting right 64 bits.

| aa19c9, 7 lines                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>using ul = uint64_t; using L = __uint128_t; struct FastMod {     ul b, m; FastMod(ul b) : b(b), m(-1ULL / b) {}     ul reduce(ul a) {         ul q = (ul)((__uint128_t(m) * a) &gt;&gt; 64), r = a - q * b;         return r - (r &gt;= b) * b; } };</pre> |

ModSqrt.h  
**Description:** Tonelli-Shanks algorithm for square roots mod a prime. -1 if doesn't exist.  
**Usage:** sqrt(mi((ll)1e10)); // 100000  
**Time:**  $\mathcal{O}(\log^2(MOD))$

| "ModInt.h"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | bcfa63, 14 lines |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <pre>using T = int; T sqrt(mi a) {     mi p = pow(a, (MOD-1)/2);     if (p.v != 1) return p.v == 0 ? 0 : -1;     T s = MOD-1; int r = 0; while (s%2 == 0) s /= 2, ++r;     mi n = 2; while (pow(n, (MOD-1)/2).v == 1) n = T(n)+1;     // n non-square, ord(g)=2^r, ord(b)=2^m, ord(g)=2^r, m&lt;r     for (mi x = pow(a, (s+1)/2), b = pow(a,s), g = pow(n,s);) {         if (b.v == 1) return min(x.v,MOD-x.v); // x^2=ab         int m = 0; for (mi t = b; t.v != 1; t *= t) ++m;         rep(r-m-1) g *= g; // ord(g)=2^{m+1}         x *= g, g *= g, b *= g, r = m; // ord(g)=2^m, ord(b)&lt;2^m     } }</pre> |                  |

ModSum.h  
**Description:** Counts  $\#$  of lattice points  $(x,y)$  in the triangle  $1 \leq x, 1 \leq y, ax+by \leq s$  (mod  $2^{64}$ ) and related quantities.  
**Time:**  $\mathcal{O}(\log ab)$

| 23cbf6, 20 lines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>using ul = uint64_t; ul sum2(ul n) { return n/2*((n-1) 1); } // sum(0..n-1) // \return  f(x,y)   1 &lt;= x, 1 &lt;= y, a*x+b*y &lt;= S  //      = sum_{i=1}^{qs} (S-a*i)/b ul triSum(ul a, ul b, ul s) { assert(a &gt; 0 &amp;&amp; b &gt; 0);     ul qs = s/a, rs = s%a; // ans = sum_{i=0}^{qs-1} (i*a+rs)/b     ul ad = a/b*sum2(qs)+rs/b*qs; a %= b, rs %= b;     return ad+(a?triSum(b,a,a*qs+rs):0); // reduce if a &gt;= b } // then swap x and y axes and recurse  // \return sum_{x=0}^{n-1} (a*x+b)/m</pre> |

```
//      = |f(x,y) | 0 < m*y <= a*x+b < a*n+b}|
// assuming a*n+b does not overflow
ul divSum(ul n, ul a, ul b, ul m) { assert(m > 0);
    ul extra = b/m*n; b %= m;
    return extra+(a?triSum(m,a,a*n+b):0); }
// \return sum_{x=0}^{n-1} (a*x+b)%m
ul modSum(ul n, ll a, ll b, ul m) { assert(m > 0);
    a = (a%m+m)%m, b = (b%m+m)%m;
    return a*sum2(n)+b*n-m*divSum(n,a,b,m); }
```

## 4.2 Primality

### 4.2.1 Primes

$p = 962592769$  is such that  $2^{21} \mid p-1$ , which may be useful. For hashing use 970592641 (31-bit number), 31443539979727 (45-bit), 3006703054056749 (52-bit). There are 78498 primes less than 1 000 000.

Primitive roots exist modulo any prime power  $p^a$ , except for  $p = 2, a > 2$ , and there are  $\phi(\phi(p^a))$  many. For  $p = 2, a > 2$ , the group  $\mathbb{Z}_{2^a}^\times$  is instead isomorphic to  $\mathbb{Z}_2 \times \mathbb{Z}_{2^{a-2}}$ .

### 4.2.2 Divisors

$\sum_{d \mid n} d = O(n \log \log n)$ .

The number of divisors of  $n$  is at most around 100 for  $n < 5e4$ , 500 for  $n < 1e7$ , 2000 for  $n < 1e10$ , 200 000 for  $n < 1e19$ .

**Dirichlet Convolution:** Given a function  $f(x)$ , let

$(f * g)(x) = \sum_{d \mid x} g(d) f(x/d)$ .

If the partial sums  $s_{f*g}(n), s_g(n)$  can be computed in  $O(1)$  and  $s_f(1 \dots n^{2/3})$  can be computed in  $O\left(n^{2/3}\right)$  then all  $s_f\left(\frac{n}{d}\right)$  can as well. Use

$$s_{f*g}(n) = \sum_{d=1}^n g(d) s_f(n/d).$$

If  $f(x) = \mu(x)$  then  $g(x) = 1, (f * g)(x) = (x == 1)$ , and  $s_f(n) = 1 - \sum_{i=2}^n s_f(n/i)$ .

If  $f(x) = \phi(x)$  then  $g(x) = 1, (f * g)(x) = x$ , and  $s_f(n) = \frac{n(n+1)}{2} - \sum_{i=2}^n s_f(n/i)$ .

FracInterval.h  
**Description:** Given fractions  $a < b$  with non-negative numerators and denominators, finds fraction  $f$  with lowest denominator such that  $a < f < b$ . Should work with all numbers less than  $2^{62}$ .

```
pl bet(pl a, pl b) {
    ll num = a.f/a.s; a.f -= num*a.s, b.f -= num*b.s;
    if (b.f > b.s) return {1+num,1};
    auto x = bet({b.s,b.f},{a.s,a.f});
    return {x.s+num*x.f,x.f};
}
```

Euclid.h  
**Description:** Generalized Euclidean algorithm. euclid and invGeneral work for  $A, B < 2^{62}$ .

**Time:**  $\mathcal{O}(\log AB)$

|                 |
|-----------------|
| c7e528, 9 lines |
|-----------------|

```
// ceil(a/b)
// ll cdiv(ll a, ll b) { return a/b+((a^b)>0&&a%b); }
pl euclid(ll A, ll B) { // For A,B>=0, finds (x,y) s.t.
    // Ax+By=gcd(A,B), |Ax|,|By|<=AB/gcd(A,B)
    if (!B) return {1,0};
    pl p = euclid(B,A%B); return {p.s,p.f-A/B*p.s}; }
ll invGeneral(ll A, ll B) { // find x in [0,B) such that Ax=1
    ↪mod B
    pl p = euclid(A,B); assert(p.f*A+p.s*B == 1);
    return p.f+(p.f<0)*B; } // must have gcd(A,B)=1
```

CRT.h  
**Description:** Chinese Remainder Theorem.  $a.f \pmod{a.s}, b.f \pmod{b.s} \implies ? \pmod{\text{lcm}(a.s,b.s)}$ . Should work for  $ab < 2^{62}$ .

| "Euclid.h"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 23df64, 10 lines |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <pre>pl CRT(pl a, pl b) { assert(0 &lt;= a.f &amp;&amp; a.f &lt; a.s &amp;&amp; 0 &lt;= b.f     ↪&amp;&amp; b.f &lt; b.s);     if (a.s &lt; b.s) swap(a,b); // will overflow if b.s^2 &gt; 2^{62}     ll x,y; tie(x,y) = euclid(a.s,b.s);     ll g = a.s*x+b.s*y, l = a.s/g*b.s;     if ((b.f-a.f)%g) return {-1,-1}; // no solution     // ?a.s+a.f \equiv b.f \pmod{b.s}     // ?=(b.f-a.f)/g*(a.s/g)^{-1} \pmod{b.s/g}     x = (b.f-a.f)%b.s*x%b.s/g*a.s+a.f;     return {x+(x&lt;0)*l,1}; }</pre> |                  |

ModArith.h  
**Description:** Statistics on mod'ed arithmetic series. minBetween and minRemainder both assume that  $0 \leq L \leq R < B, AB < 2^{62}$

| f68a6d, 40 lines                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>ll minBetween(ll A, ll B, ll L, ll R) {     // min x s.t. exists y s.t. L &lt;= A*x-B*y &lt;= R     A %= B;     if (L == 0) return 0;     if (A == 0) return -1;     ll k = cdiv(L,A); if (A*k &lt;= R) return k;     ll x = minBetween(B,A,A-R%A,A-L%A); // min x s.t. exists y     // s.t. -R &lt;= Bx-Ay &lt;= -L     return x == -1 ? x : cdiv(B*x+L,A); // solve for y }</pre> |

```
// find min((Ax+C)%B) for 0 <= x <= M
// aka find minimum non-negative value of A*x-B*y+C
// where 0 <= x <= M, 0 <= y
ll minRemainder(ll A, ll B, ll C, ll M) {
    assert(A >= 0 && B > 0 && C >= 0 && M >= 0);
    A %= B, C %= B; ckmin(M,B-1);
    if (A == 0) return C;
    if (C >= A) { // make sure C<A
        ll ad = cdiv(B-C,A);
        M -= ad; if (M < 0) return C;
        C += ad*A-B;
    }
    ll q = B/A, new_B = B%A; // new_B < A
    if (new_B == 0) return C; // B=q*A
```

```
// now minimize A*x-new_B*y+C
// where 0 <= x,y and x+q*y <= M, 0 <= C < new_B < A
// q*y -> C-new_B*y
if (C/new_B > M/q) return C-M/q*new_B;
M -= C/new_B*q; C %= new_B; // now C < new_B

// given y, we can compute x = ceil[((B-q*A)*y-C)/A]
// so x+q*y = ceil((B*y-C)/A) <= M
ll max_Y = (M+A+C)/B; // must have y <= max_Y
ll max_X = cdiv(new_B*max_Y-C,A); // must have x <= max_X
```

```
if (max_X*A-new_B*max_Y+C >= new_B) --max_X;
// now we can remove upper bound on y
return minRemainder(A,new_B,C,max_X);
}
```

4.3 Pythagorean Triples

The Pythagorean triples are uniquely generated by

$a = k \cdot (m^2 - n^2), \quad b = k \cdot (2mn), \quad c = k \cdot (m^2 + n^2),$

with  $m > n > 0, k > 0, m \perp n$ , and either  $m$  or  $n$  even.

4.4 Lifting the Exponent

For  $n > 0, p$  prime, and ints  $x, y$  s.t.  $p \nmid x, y$  and  $p | x - y$ :

- $p \neq 2$  or  $p = 2, 4 | x - y \implies v_p(x^n - y^n) = v_p(x - y) + v_p(n).$
- $p = 2, 2 | n \implies v_2(x^n - y^n) = v_2((x^2)^{n/2} - (y^2)^{n/2}).$

Combinatorial (5)

5.1 Permutations

5.1.1 Cycles

Let  $g_S(n)$  be the number of  $n$ -permutations whose cycle lengths all belong to the set  $S$ . Then

$$\sum_{n=0}^\infty g_S(n) \frac{x^n}{n!} = \exp\left(\sum_{n \in S} \frac{x^n}{n}\right)$$

5.1.2 Burnside’s lemma

Given a group  $G$  of symmetries and a set  $X$ , the number of elements of  $X$  up to symmetry equals

$$\frac{1}{|G|} \sum_{g \in G} |X^g|,$$

where  $X^g$  are the elements fixed by  $g$  ( $g.x = x$ ).

If  $f(n)$  counts “configurations” (of some sort) of length  $n$ , we can ignore rotational symmetry using  $G = \mathbb{Z}_n$  to get

$$g(n) = \frac{1}{n} \sum_{k=0}^{n-1} f(\gcd(n, k)) = \frac{1}{n} \sum_{k | n} f(k) \phi(n/k).$$

5.2 Partitions and subsets

5.2.1 Partition function

Number of ways of writing  $n$  as a sum of positive integers, disregarding the order of the summands.

$$p(0) = 1, \quad p(n) = \sum_{k \in \mathbb{Z} \setminus \{0\}} (-1)^{k+1} p(n - k(3k - 1)/2)$$

$$p(n) \sim 0.145/n \cdot \exp(2.56\sqrt{n})$$

|        |   |   |   |   |   |   |    |    |    |    |     |            |            |
|--------|---|---|---|---|---|---|----|----|----|----|-----|------------|------------|
| $n$    | 0 | 1 | 2 | 3 | 4 | 5 | 6  | 7  | 8  | 9  | 20  | 50         | 100        |
| $p(n)$ | 1 | 1 | 2 | 3 | 5 | 7 | 11 | 15 | 22 | 30 | 627 | $\sim 2e5$ | $\sim 2e8$ |

5.2.2 Lucas’ Theorem

Let  $n, m$  be non-negative integers and  $p$  a prime. Write  $n = n_k p^k + \dots + n_1 p + n_0$  and  $m = m_k p^k + \dots + m_1 p + m_0$ . Then  $\binom{n}{m} \equiv \prod_{i=0}^k \binom{n_i}{m_i} \pmod{p}$ .

5.3 General purpose numbers

5.3.1 Bernoulli numbers

EGF of Bernoulli numbers is  $B(t) = \frac{t}{e^t - 1}$  (FFT-able).

$$B[0, \dots] = [1, -\frac{1}{2}, \frac{1}{6}, 0, -\frac{1}{30}, 0, \frac{1}{42}, \dots]$$

Sums of powers:

$$\sum_{i=1}^n i^m = \frac{1}{m+1} \sum_{k=0}^m \binom{m+1}{k} B_k (n+1)^{m+1-k}$$

Euler-Maclaurin formula for infinite sums:

$$\begin{aligned} \sum_{i=m}^\infty f(i) &= \int_m^\infty f(x) dx - \sum_{k=1}^\infty \frac{B_k}{k!} f^{(k-1)}(m) \\ &\approx \int_m^\infty f(x) dx + \frac{f(m)}{2} - \frac{f'(m)}{12} + \frac{f'''(m)}{720} + O(f^{(5)}(m)) \end{aligned}$$

5.3.2 Stirling numbers of the first kind

Number of permutations on  $n$  items with  $k$  cycles.

$$\begin{aligned} c(n, k) &= c(n-1, k-1) + (n-1)c(n-1, k), \quad c(0, 0) = 1 \\ \sum_{k=0}^n c(n, k) x^k &= x(x+1) \dots (x+n-1) \end{aligned}$$

$$\begin{aligned} c(8, k) &= 8, 0, 5040, 13068, 13132, 6769, 1960, 322, 28, 1 \\ c(n, 2) &= 0, 0, 1, 3, 11, 50, 274, 1764, 13068, 109584, \dots \end{aligned}$$

5.3.3 Eulerian numbers

Number of permutations  $\pi \in S_n$  in which exactly  $k$  elements are greater than the previous element.  $k$   $j$ :s s.t.  $\pi(j) > \pi(j+1)$ ,  $k+1$   $j$ :s s.t.  $\pi(j) \geq j$ ,  $k$   $j$ :s s.t.  $\pi(j) > j$ .

$$E(n, k) = (n - k)E(n - 1, k - 1) + (k + 1)E(n - 1, k)$$

$$E(n, 0) = E(n, n - 1) = 1$$

$$E(n, k) = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n$$

5.3.4 Stirling numbers of the second kind

Partitions of  $n$  distinct elements into exactly  $k$  groups.

$$S(n, k) = S(n - 1, k - 1) + kS(n - 1, k)$$

$$S(n, 1) = S(n, n) = 1$$

$$S(n, k) = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$$

5.3.5 Bell numbers

Total number of partitions of  $n$  distinct elements.  $B(n) = 1, 1, 2, 5, 15, 52, 203, 877, 4140, 21147, \dots$ . For  $p$  prime,

$$B(p^m + n) \equiv mB(n) + B(n + 1) \pmod{p}$$

5.3.6 Labeled unrooted trees

- # on  $n$  vertices:  $n^{n-2}$
- # on  $k$  existing trees of size  $n_i$ :  $n_1 n_2 \dots n_k n^{k-2}$
- # with degrees  $d_i$ :  $(n-2)! / ((d_1-1)! \dots (d_n-1)!)$

5.3.7 Catalan numbers

$$C_n = \frac{1}{n+1} \binom{2n}{n} = \binom{2n}{n} - \binom{2n}{n+1} = \frac{(2n)!}{(n+1)!n!}$$

$$C_0 = 1, \quad C_{n+1} = \frac{2(2n+1)}{n+2} C_n, \quad C_{n+1} = \sum C_i C_{n-i}$$

$$C_n = 1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, 16796, 58786, \dots$$

- sub-diagonal monotone paths in an  $n \times n$  grid.
- strings with  $n$  pairs of parenthesis, correctly nested.
- binary trees with  $n+1$  leaves (0 or 2 children).
- ordered trees with  $n+1$  vertices.
- ways a convex polygon with  $n+2$  sides can be cut into triangles by connecting vertices with straight lines.
- permutations of  $[n]$  with no 3-term increasing subseq.

5.4 Young Tableaux

Let a **Young diagram** have shape  $\lambda = (\lambda_1 \geq \dots \geq \lambda_k)$ , where  $\lambda_i$  equals the number of cells in the  $i$ -th (left-justified) row from the top. A **Young tableau** of shape  $\lambda$  is a filling of the  $n = \sum \lambda_i$  cells with a permutation of  $1 \dots n$  such that each row and column is increasing.

**Hook-Length Formula:** For the cell in position  $(i, j)$ , let  $h_\lambda(i, j) = |\{(I, J) | i \leq I, j \leq J, (I = i \text{ or } J = j)\}|$ . The number of Young tableaux of shape  $\lambda$  is equal to  $f^\lambda = \frac{n!}{\prod h_\lambda(i, j)}$ .

**Schensted’s Algorithm:** converts a permutation  $\sigma$  of length  $n$  into a pair of Young Tableaux  $(S(\sigma), T(\sigma))$  of the same shape. When inserting  $x = \sigma_i$ ,

1. Add  $x$  to the first row of  $S$  by inserting  $x$  in place of the largest  $y$  with  $x < y$ . If  $y$  doesn’t exist, push  $x$  to the end of the row, set the value of  $T$  at that position to be  $i$ , and stop.
2. Add  $y$  to the second row using the same rule, keep repeating as necessary.

All pairs  $(S(\sigma), T(\sigma))$  of the same shape correspond to a unique  $\sigma$ , so  $n! = \sum (f^\lambda)^2$ . Also,  $S(\sigma^R) = S(\sigma)^T$ .

Let  $d_k(\sigma), a_k(\sigma)$  be the lengths of the longest subseqs which are a union of  $k$  decreasing/ascending subseqs, respectively. Then  $a_k(\sigma) = \sum_{i=1}^k \lambda_i, d_k(\sigma) = \sum_{i=1}^k \lambda_i^*$ , where  $\lambda_i^*$  is size of the  $i$ -th column.

5.5 Other

DeBruijnSeq.h  
**Description:** Given alphabet  $[0, k)$  constructs a cyclic string of length  $k^n$  that contains every length  $n$  string as substr.

```

a6961b, 13 lines
vi deBruijnSeq(int k, int n) {
    if (k == 1) return {};
    vi seq, aux(n+1);
    function<void(int,int)> gen = [&](int t, int p) {
        if (t > n) { // +lyndon word of len p
            if (n%p == 0) FOR(i,1,p+1) seq.pb(aux[i]);
        } else {
            aux[t] = aux[t-p]; gen(t+1,p);
            while (++aux[t] < k) gen(t+1,t);
        }
    };
    gen(1,1); return seq;
}
```

NimProduct.h

**Description:** Product of nimbers is associative, commutative, and distributive over addition (xor). Forms finite field of size  $2^{2^k}$ . Defined by  $ab = \text{mex}(\{a'b + ab' + a'b' : a' < a, b' < b\})$ . Application: Given 1D coin turning games  $G_1, G_2$   $G_1 \times G_2$  is the 2D coin turning game defined as follows. If turning coins at  $x_1, x_2, \dots, x_m$  is legal in  $G_1$  and  $y_1, y_2, \dots, y_n$  is legal in  $G_2$ , then turning coins at all positions  $(x_i, y_j)$  is legal assuming that the coin at  $(x_m, y_n)$  goes from heads to tails. Then the grundy function  $g(x, y)$  of  $G_1 \times G_2$  is  $g_1(x) \times g_2(y)$ .  
**Time:**  $64^2$  xors per multiplication, memorize to speed up.

```

5afe17, 46 lines
using ul = uint64_t;
struct Precalc {
    ul tmp[64][64], y[8][8][256];
    unsigned char x[256][256];
    Precalc() { // small nim products, all < 256
        FOR(i,256) FOR(j,256) x[i][j] = mult<8>(i,j);
        FOR(i,8) FOR(j,i+1) FOR(k,256)
            y[i][j][k] = mult<64>(prod2(8*i,8*j),k);
    }
    ul prod2(int i, int j) { // nim prod of 2^i, 2^j
        ul& u = tmp[i][j]; if (u) return u;
        if (!(i&j)) return u = 1ULL<<(i|j);
        int a = (i&j)&~(i&j); // a=2^k, consider 2^{2^k}
        return u=prod2(i^a,j)^prod2((i^a)|(a-1),(j^a)|(i&(a-1)));
        // 2^{2^k}*2^{2^k} = 2^{2^k}+2^{2^k-1}
    } // 2^{2^i}*2^{2^j} = 2^{2^i+2^j} if i<j
    template<int L> ul mult(ul a, ul b) {
        ul c = 0; FOR(i,L) if (a>>i&1)
            FOR(j,L) if (b>>j&1) c ^= prod2(i,j);
        return c;
    }
    // 2^{8*i}*(>>(8*i)&255) * 2^{8*j}*(>>(8*j)&255)
    // -> (2^{8*i}*2^{8*j})*((>>(8*i)&255)*(>>(8*j)&255))
    ul multFast(ul a, ul b) const { // faster nim product
        ul res = 0; auto f=[](ul c,int d){return c>>(8*d)&255;};
        FOR(i,8) {
            FOR(j,i) res ^= y[i][j][x[f(a,i)][f(b,j)]]
                ^x[f(a,j)][f(b,i)];
            res ^= y[i][i][x[f(a,i)][f(b,i)]];
        }
        return res;
    }
};
```

```

const Precalc P;

struct nb { // nimer
    ul x; nb() { x = 0; };
    nb(ul _x): x(_x) {}
    explicit operator ul() { return x; }
    nb operator+(nb y) { return nb(x^y.x); }
    nb operator*(nb y) { return nb(P.multFast(x,y.x)); }
    friend nb pow(nb b, ul p) {
        nb res = 1; for (;p;p/=2,b=b*b) if (p&1) res = res*b;
        return res; } // b^{2^{2^A}-1}=1 where 2^{2^A} > b
    friend nb inv(nb b) { return pow(b,-2); }
};
```

MatroidIsect.h

**Description:** Computes a set of maximum size which is independent in both graphic and colorful matroids, aka a spanning forest where no two edges are of the same color. In general, construct the exchange graph and find a shortest path. Can apply similar concept to partition matroid.  
**Usage:** MatroidIsect<Gmat,Cmat> M(sz(ed),Gmat(ed),Cmat(col))  
**Time:**  $\mathcal{O}(GI^{1.5})$  calls to oracles, where  $G$  is size of ground set and  $I$  is size of independent set.

```

d0051c, 51 lines
"DSU.h"
struct Gmat { // graphic matroid
    int V = 0; vpi ed; DSU D;
    Gmat(vpi _ed):ed(_ed) {
        map<int,int> m; each(t,ed) m[t.f] = m[t.s] = 0;
        each(t,m) t.s = V++;
        each(t,ed) t.f = m[t.f], t.s = m[t.s];
    }
    void clear() { D.init(V); }
    void ins(int i) { assert(D.unite(ed[i].f,ed[i].s)); }
    bool indep(int i) { return !D.sameSet(ed[i].f,ed[i].s); }
};
struct Cmat { // colorful matroid
    int C = 0; vi col; V<bool> used;
    Cmat(vi col):col(col) {each(t,col) ckmax(C,t+1); }
    void clear() { used.assign(C,0); }
    void ins(int i) { used[col[i]] = 1; }
    bool indep(int i) { return !used[col[i]]; }
};
template<class M1, class M2> struct MatroidIsect {
    int n; V<bool> iset; M1 m1; M2 m2;
    bool augment() {
        vi pre(n+1,-1); queue<int> q({n});
        while (sz(q)) {
            int x = q.ft; q.pop();
            if (iset[x]) {
                m1.clear(); FOR(i,n) if (iset[i] && i != x) m1.ins(i);
                FOR(i,n) if (!iset[i] && pre[i] == -1 && m1.indep(i))
                    pre[i] = x, q.push(i);
            } else {
                auto backE = [&]() { // back edge
                    m2.clear();
                    FOR(c,2)FOR(i,n)if((x==i||iset[i])&&(pre[i]==-1)==c){
                        if (!m2.indep(i))return c?pre[i]=x,q.push(i),i:-1;
                        m2.ins(i); }
                    return n;
                };
                for (int y; (y = backE()) != -1;) if (y == n) {
                    for(; x != n; x = pre[x]) iset[x] = !iset[x];
                    return 1; }
            }
        }
        return 0;
    }
    MatroidIsect(int n, M1 m1, M2 m2):n(n), m1(m1), m2(m2) {
        iset.assign(n+1,0); iset[n] = 1;
        m1.clear(); m2.clear(); // greedily add to basis
    }
};
```

```

R0F(i,n) if (m1.indep(i) && m2.indep(i))
    iset[i] = 1, m1.ins(i), m2.ins(i);
while (augment());
}
};
```

Numerical (6)

6.1 Matrix

```

Matrix.h
Description: 2D matrix operations.
b18e29, 21 lines
"ModInt.h"
using T = mi;
using Mat = V<V<T>>; // use array instead if tight TL

Mat makeMat(int r, int c) { return Mat(r,V<T>(c)); }
Mat makeId(int n) {
    Mat m = makeMat(n,n); FOR(i,n) m[i][i] = 1;
    return m;
}
Mat operator*(const Mat& a, const Mat& b) {
    int x = sz(a), y = sz[a[0]], z = sz[b[0]];
    assert(y == sz(b)); Mat c = makeMat(x,z);
    FOR(i,x) FOR(j,y) FOR(k,z) c[i][k] += a[i][j]*b[j][k];
    return c;
}
Mat& operator*=(Mat& a, const Mat& b) { return a = a*b; }
Mat pow(Mat m, ll p) {
    int n = sz(m); assert(n == sz(m[0]) && p >= 0);
    Mat res = makeId(n);
    for (; p; p /= 2, m *= m) if (p&1) res *= m;
    return res;
}
```

MatrixInv.h

**Description:** Uses gaussian elimination to convert into reduced row echelon form and calculates determinant. For determinant via arbitrary modulus, use a modified form of the Euclidean algorithm because modular inverse may not exist. If you have computed  $A^{-1} \pmod{p^k}$ , then the inverse  $\pmod{p^{2k}}$  is  $A^{-1}(2I - AA^{-1})$ .  
**Time:**  $\mathcal{O}(N^3)$ , determinant of  $1000 \times 1000$  matrix of modints in 1 second if you reduce # of operations by half

```

73ec43, 38 lines
const db EPS = 1e-9; // adjust?
int getRow(V<V<db>>& m, int R, int i, int nex) {
    pair<db,int> bes{0,-1}; // find row with max abs value
    FOR(j,nex,R) ckmax(bes,{abs(m[j][i]),j});
    return bes.f < EPS ? -1 : bes.s; }
int getRow(V<vmi>& m, int R, int i, int nex) {
    FOR(j,nex,R) if (m[j][i] != 0) return j;
    return -1; }
pair<T,int> gauss(Mat& m) { // convert to reduced row echelon
    ⇐form
    if (!sz(m)) return {1,0};
    int R = sz(m), C = sz(m[0]), rank = 0, nex = 0;
    T prod = 1; // determinant
    FOR(i,C) {
        int row = getRow(m,R,i,nex);
        if (row == -1) { prod = 0; continue; }
        if (row != nex) prod *= -1, swap(m[row],m[nex]);
        prod *= m[nex][i]; rank++;
        T x = 1/m[nex][i]; FOR(k,i,C) m[nex][k] *= x;
        FOR(j,R) if (j != nex) {
            T v = m[j][i]; if (v == 0) continue;
            FOR(k,i,C) m[j][k] -= v*m[nex][k];
        }
        nex++;
    }
};
```

```
    }
    return {prod,rank};
}
Mat inv(Mat m) {
    int R = sz(m); assert(R == sz(m[0]));
    Mat x = makeMat(R,2*R);
    F0R(i,R) {
        x[i][i+R] = 1;
        F0R(j,R) x[i][j] = m[i][j];
    }
    if (gauss(x).s != R) return Mat();
    Mat res = makeMat(R,R);
    F0R(i,R) F0R(j,R) res[i][j] = x[i][j+R];
    return res;
}
```

**MatrixTree.h**  
**Description:** Kirchhoff's Matrix Tree Theorem. Given adjacency matrix, calculates # of spanning trees.

|               |                  |
|---------------|------------------|
| "MatrixInv.h" | 48363d, 11 lines |
|---------------|------------------|

```
T numSpan(const Mat& m) {
    int n = sz(m); Mat res = makeMat(n-1,n-1);
    F0R(i,n) FOR(j,i+1,n) {
        mi ed = m[i][j]; res[i][i] += ed;
        if (j != n-1) {
            res[j][j] += ed;
            res[i][j] -= ed, res[j][i] -= ed;
        }
    }
    return gauss(res).f;
}
```

**ShermanMorrison.h**  
**Description:** Calculates  $(A + uv^T)^{-1}$  given  $B = A^{-1}$ . Not invertible if sum=0.

|               |                 |
|---------------|-----------------|
| "MatrixInv.h" | 3a3f34, 7 lines |
|---------------|-----------------|

```
void ad(Mat& B, const V<T>& u, const V<T>& v) {
    int n = sz(A); V<T> x(n), y(n);
    F0R(i,n) F0R(j,n)
        x[i] += B[i][j]*u[j], y[j] += v[i]*B[i][j];
    T sum = 1; F0R(i,n) F0R(j,n) sum += v[i]*B[i][j]*u[j];
    F0R(i,n) F0R(j,n) B[i][j] -= x[i]*y[j]/sum;
}
```

## 6.2 Polynomials

**Poly.h**  
**Description:** Basic poly ops including division. Can replace T with double, complex.

|            |                  |
|------------|------------------|
| "ModInt.h" | cd218a, 73 lines |
|------------|------------------|

```
using T = mi; using poly = V<T>;
void remz(poly& p) { while (sz(p)&&p.bk==T(0)) p.pop_back(); }
poly REMZ(poly p) { remz(p); return p; }
poly rev(poly p) { reverse(all(p)); return p; }
poly shift(poly p, int x) {
    if (x >= 0) p.insert(begin(p),x,0);
    else assert(sz(p)+x >= 0), p.erase(begin(p),begin(p)-x);
    return p;
}
poly RSZ(const poly& p, int x) {
    if (x <= sz(p)) return poly(begin(p),begin(p)+x);
    poly q = p; q.rsz(x); return q; }
T eval(const poly& p, T x) { // evaluate at point x
    T res = 0; R0F(i,sz(p)) res = x*res+p[i];
    return res; }
poly dif(const poly& p) { // differentiate
    poly res; FOR(i,1,sz(p)) res.pb(T(i)*p[i]);
    return res; }
```

```
poly integ(const poly& p) { // integrate
    static poly invs{0,1};
    for (int i = sz(invs); i <= sz(p); ++i)
        invs.pb(-MOD/i*invs[MOD%i]);
    poly res(sz(p)+1); F0R(i,sz(p)) res[i+1] = p[i]*invs[i+1];
    return res;
}
```

```
poly& operator+=(poly& l, const poly& r) {
    l.rsz(max(sz(l),sz(r))); F0R(i,sz(r)) l[i] += r[i];
    return l; }
poly& operator-=(poly& l, const poly& r) {
    l.rsz(max(sz(l),sz(r))); F0R(i,sz(r)) l[i] -= r[i];
    return l; }
poly& operator*=(poly& l, const T& r) { each(t,l) t *= r;
    return l; }
poly& operator/=(poly& l, const T& r) { each(t,l) t /= r;
    return l; }
poly operator+(poly l, const poly& r) { return l += r; }
poly operator-(poly l, const poly& r) { return l -= r; }
poly operator*(poly l, const poly& r) { return l *= r; }
poly operator/(poly l, const poly& r) { return l /= r; }
poly operator*(const poly& l, const poly& r) {
    if (!min(sz(l),sz(r))) return {};
    poly x(sz(l)+sz(r)-1);
    F0R(i,sz(l)) F0R(j,sz(r)) x[i+j] += l[i]*r[j];
    return x;
}
poly& operator*=(poly& l, const poly& r) { return l = l*r; }
```

```
pair<poly,poly> quoRemSlow(poly a, poly b) {
    remz(a); remz(b); assert(sz(b));
    T lst = b.bk, B = T(1)/lst; each(t,a) t *= B;
    each(t,b) t *= B;
    poly q(max(sz(a)-sz(b)+1,0));
    for (int dif; (dif=sz(a)-sz(b)) >= 0; remz(a)) {
        q[dif] = a.bk; F0R(i,sz(b)) a[i+dif] -= q[dif]*b[i]; }
    each(t,a) t *= lst;
    return {q,a}; // quotient, remainder
}
poly operator%(const poly& a, const poly& b) {
    return quoRemSlow(a,b).s; }
T resultant(poly a, poly b) { // R(A,B)
    // =b_m^n*prod_{j=1}^mA(mu_j)
    // =b_m^na_n^m*prod_{i=1}^nprod_{j=1}^m(mu_j-lambda_i)
    // =(-1)^{mn}a_n^m*prod_{i=1}^nB(lambda_i)
    // =(-1)^{nm}R(B,A)
    // Also, R(A,B)=b_m^f(deg(A)-deg(A-CB))R(A-CB,B)
    int ad = sz(a)-1, bd = sz(b)-1;
    if (bd <= 0) return bd < 0 ? 0 : pow(b.bk,ad);
    int pw = ad; a = a%b; pw -= (ad = sz(a)-1);
    return resultant(b,a)*pow(b.bk,pw)*T((bd&ad&1)?-1:1);
}
```

**PolyInterpolate.h**  
**Description:**  $n$  points determine unique polynomial of degree  $\leq n-1$ . For numerical precision pick  $v[k].f = c * \cos(k/(n-1) * \pi), k = 0 \dots n-1$ .  
**Time:**  $\mathcal{O}(n^2)$

|          |                 |
|----------|-----------------|
| "Poly.h" | aada3a, 8 lines |
|----------|-----------------|

```
poly interpolate(V<pair<T,T>> v) {
    poly res, tmp{1};
    F0R(i,sz(v)) { T prod = 1; // add one point at a time
        F0R(j,i) v[i].s -= prod*v[j].s, prod *= v[i].f-v[j].f;
        v[i].s /= prod; res += v[i].s*tmp; tmp *= poly{-v[i].f,1};
    } // add multiple of (x-v[0].f)*(x-v[1].f)*...*(x-v[i-1].f)
    return res;
}
```

```
FFT.h
Description: Multiply polynomials of ints for any modulus < 231. For XOR convolution ignore m within fft.
Time:  $\mathcal{O}(N \log N)$ . For  $N = 10^6$ , conv ~0.13ms, conv-general ~320ms.
"ModInt.h" 19ab20, 39 lines
// const int MOD = 998244353;
tcT> void fft(V<T>& A, bool invert = 0) { // NTT
    int n = sz(A); assert((T::mod-1)%n == 0); V<T> B(n);
    for(int b = n/2; b /= 2, swap(A,B)) { // w = n/b^th root
        T w = pow(T::rt(), (T::mod-1)/n*b), m = 1;
        for(int i = 0; i < n; i += b*2, m *= w) F0R(j,b) {
            T u = A[i+j], v = A[i+j+b]*m;
            B[i/2+j] = u+v; B[i/2+j+n/2] = u-v;
        }
    }
    if (invert) { reverse(l+all(A));
        T z = inv(T(n)); each(t,A) t *= z; }
} // for NTT-able moduli
tcT> V<T> conv(V<T> A, V<T> B) {
    if (!min(sz(A),sz(B))) return {};
    int s = sz(A)+sz(B)-1, n = 1; for (; n < s; n *= 2);
    A.rsz(n), fft(A); B.rsz(n), fft(B);
    F0R(i,n) A[i] *= B[i];
    fft(A,1); A.rsz(s); return A;
}
template<class M, class T> V<M> mulMod(const V<T>& x, const V<T
    ↪& y) {
    auto con = [] (const V<T>& v) {
        V<M> w(sz(v)); F0R(i,sz(v)) w[i] = (int)v[i];
        return w; };
    return conv(con(x), con(y));
} // arbitrary moduli
tcT> V<T> conv_general(const V<T>& A, const V<T>& B) {
    using m0 = mint<(119<<23)+1,62>; auto c0 = mulMod<m0>(A,B);
    using m1 = mint<(5<<25)+1, 62>; auto c1 = mulMod<m1>(A,B);
    using m2 = mint<(7<<26)+1, 62>; auto c2 = mulMod<m2>(A,B);
    int n = sz(c0); V<T> res(n); m1 r01 = inv(m1(m0::mod));
    m2 r02 = inv(m2(m0::mod)), r12 = inv(m2(m1::mod));
    F0R(i,n) { // a=remainder mod m0::mod, b fixes it mod m1::mod
        int a = c0[i].v, b = ((c1[i]-a)*r01).v,
            c = ((c2[i]-a)*r02-b)*r12.v;
        res[i] = (T(c)*m1::mod+b)*m0::mod+a; // c fixes m2::mod
    }
    return res;
}
```

**PolyInvSimpler.h**  
**Description:** computes  $A^{-1}$  such that  $AA^{-1} \equiv 1 \pmod{x^n}$ . Newton's method: If you want  $F(x) = 0$  and  $F(Q_k) \equiv 0 \pmod{x^a}$  then  $Q_{k+1} = Q_k - \frac{F(Q_k)}{F'(Q_k)} \pmod{x^{2a}}$  satisfies  $F(Q_{k+1}) \equiv 0 \pmod{x^{2a}}$ . Application: if  $f(n), g(n)$  are the #s of forests and trees on  $n$  nodes then  $\sum_{n=0}^{\infty} f(n)x^n = \exp\left(\sum_{n=1}^{\infty} \frac{g(n)}{n!}\right)$ .

**Usage:** vmi v{1,5,2,3,4}; ps(exp(2\*log(v,9),9)); // squares v  
**Time:**  $\mathcal{O}(N \log N)$ . For  $N = 5 \cdot 10^5$ , inv~270ms, log ~350ms, exp~550ms

|                   |                  |
|-------------------|------------------|
| "FFT.h", "Poly.h" | 6e5362, 30 lines |
|-------------------|------------------|

```
poly inv(poly A, int n) { // Q-(1/Q-A)/(-Q^-{-2})
    poly B{inv(A[0])};
    for (int x = 2; x/2 < n; x *= 2)
        B = 2*B-RSZ(conv(RSZ(A,x),conv(B,B)),x);
    return RSZ(B,n);
}
poly sqrt(const poly& A, int n) { // Q-(Q^2-A)/(2Q)
    assert(A[0].v == 1); poly B{1};
    for (int x = 2; x/2 < n; x *= 2)
        B = inv(T(2))*RSZ(B+conv(RSZ(A,x),inv(B,x)),x);
    return RSZ(B,n);
}
```



```
// return {quotient, remainder}
pair<poly,poly> quoRem(const poly& f, const poly& g) {
    if (sz(f) < sz(g)) return {{},f};
    poly q = conv(inv(rev(g),sz(f)-sz(g)+1),rev(f));
    q = rev(RSZ(q,sz(f)-sz(g)+1));
    poly r = RSZ(f-conv(q,g),sz(g)-1); return {q,r};
}

poly log(poly A, int n) { assert(A[0].v == 1); // (ln A)' = A'/A
    A.rsz(n); return integ(RSZ(conv(dif(A),inv(A,n-1)),n-1)); }
poly exp(poly A, int n) { assert(A[0].v == 0);
    poly B{1}, IB{1}; // inverse of B
    for (int x = 1; x < n; x *= 2) {
        IB = 2*IB-RSZ(conv(B,conv(IB,IB)),x);
        poly Q = dif(RSZ(A,x)); Q += RSZ(conv(IB,dif(B)-conv(B,Q))
            ↪,2*x-1);
        B = B+RSZ(conv(B,RSZ(A,2*x)-integ(Q)),2*x);
    }
    return RSZ(B,n);
}
```

6.3 Misc

LinearRecurrence.h  
**Description:** Berlekamp-Massey. Computes linear recurrence C of order N for sequence s of 2N terms. C[0] = 1 and for all i ≥ sz(C) − 1,  $\sum_{j=0}^{sz(C)-1} C[j]s[i-j] = 0$ .  
**Usage:** LinRec L; L.init({0,1,1,2,3}); L.eval(5); L.eval(6); // 5, 8  
**Time:**  $\text{init} \Rightarrow \mathcal{O}(N|C|)$ ,  $\text{eval} \Rightarrow \mathcal{O}(|C|^2 \log p)$  or faster with FFT

```
"Poly.h"
39ea71, 29 lines

struct LinRec {
    poly s, C, rC;
    void BM() {
        int x = 0; T b = 1;
        poly B; B = C = {1}; // B is fail vector
        F0R(i,sz(s)) { // update C after adding a term of s
            ++x; int L = sz(C), M = i+3-L;
            T d = 0; F0R(j,L) d += C[j]*s[i-j]; // [D^i]C*s
            if (d.v == 0) continue; // [D^i]C*s=0
            poly _C = C; T coef = d*inv(b);
            C.rsz(max(L,M)); F0R(j,sz(B)) C[j+x] -= coef*B[j];
            if (L < M) B = _C, b = d, x = 0;
        }
    }
    void init(const poly& _s) {
        s = _s; BM();
        rC = C; reverse(all(rC));
        C.erase(begin(C)); each(t,C) t *= -1;
    } // now s[i]=sum_{j=0}^{sz(C)-1}C[j]*s[i-j-1]
    poly getPow(ll p) { // get x^p mod rC
        if (p == 0) return {1};
        poly r = getPow(p/2); r = (r*r)%rC;
        return p&1?(r*poly{0,1})%rC:r;
    }
    T dot(poly v) { // dot product with s
        T ans = 0; F0R(i,sz(v)) ans += v[i]*s[i];
        return ans; } // get p-th term of rec
    T eval(ll p) { assert(p >= 0); return dot(getPow(p)); }
};
```

Integrate.h  
**Description:** Integration of a function over an interval using Simpson’s rule, exact for polynomials of degree up to 3. The error should be proportional to  $dif^4$ , although in practice you will want to verify that the result is stable to desired precision when epsilon changes.  
**Usage:** quad([])(db x) { return x\*x+3\*x+1; }, 2, 3) // 14.833

```
template<class F> db quad(F f, db a, db b) {
3ebaab, 6 lines
```

```
const int n = 1000;
db dif = (b-a)/2/n, tot = f(a)+f(b);
FOR(i,1,2*n) tot += f(a+i*dif)*(i&1?4:2);
return tot*dif/3;
}
```

IntegrateAdaptive.h  
**Description:** Unused. Fast integration using adaptive Simpson’s rule, exact for polynomials of degree up to 5.  
**Usage:** db z, y;  
db h(db x) { return x\*x + y\*y + z\*z <= 1; }  
db g(db y) { ::y = y; return quad(h, -1, 1); }  
db f(db z) { ::z = z; return quad(g, -1, 1); }  
db sphereVol = quad(f,-1,1), pi = sphereVol\*3/4;

```
template<class F> db simpson(F f, db a, db b) {
    db c = (a+b)/2; return (f(a)+4*f(c)+f(b))*(b-a)/6; }
template<class F> db rec(F& f, db a, db b, db eps, db S) {
    db c = (a+b)/2;
    db S1 = simpson(f,a,c), S2 = simpson(f,c,b), T = S1+S2;
    if (abs(T-S)<=15*eps || b-a<1e-10) return T+(T-S)/15;
    return rec(f,a,c,eps/2,S1)+rec(f,c,b,eps/2,S2);
}

template<class F> db quad(F f, db a, db b, db eps = 1e-8) {
    return rec(f,a,b,eps,simpson(f,a,b)); }
```

Simplex.h  
**Description:** Solves a general linear maximization problem: maximize  $c^T x$  subject to  $Ax \leq b, x \geq 0$ . Returns -inf if there is no solution, inf if there are arbitrarily good solutions, or the maximum value of  $c^T x$  otherwise. The input vector is set to an optimal x (or in the unbounded case, an arbitrary solution fulfilling the constraints). Numerical stability is not guaranteed. For better performance, define variables such that  $x = 0$  is viable.  
**Usage:** vvd A{{1,-1}, {-1,1}, {-1,-2}};  
vvd b{1,1,-4}, c{-1,-1}, x;  
T val = LPSolver(A, b, c).solve(x);  
**Time:**  $\mathcal{O}(NM \cdot \#pivots)$ , where a pivot may be e.g. an edge relaxation.

```
 $\mathcal{O}(2^N)$  in the general case.
c99f9c, 67 lines

using T = db; // double probably suffices
using vd = V<T>; using vvd = V<vd>;
const T eps = 1e-8, inf = 1./0;

#define ltj(X) if (s==-1 || mp(X[j],N[j])<mp(X[s],N[s])) s=j
struct LPSolver {
    int m, n; // # m = constraints, # n = variables
    vi N, B; // N[j] = non-basic variable (j-th column), = 0
    vvd D; // B[j] = basic variable (j-th row)
    LPSolver(const vvd& A, const vd& b, const vvd& c) :
        m(sz(b)), n(sz(c)), N(n+1), B(m), D(m+2, vd(n+2)) {
        F0R(i,m) F0R(j,n) D[i][j] = A[i][j];
        F0R(i,m) B[i] = n+i, D[i][n] = -1, D[i][n+1] = b[i];
        // B[i]: basic variable for each constraint
        // D[i][n]: artificial variable for testing feasibility
        F0R(j,n) N[j] = j, D[m][j] = -c[j];
        // D[m] stores negation of objective,
        // which we want to minimize
        N[n] = -1; D[m+1][n] = 1; // to find initial feasible
    } // solution, minimize artificial variable
    void pivot(int r, int s) { // swap B[r] (row)
        T inv = 1/D[r][s]; // with N[r] (column)
        F0R(i,m+2) if (i != r && abs(D[i][s]) > eps) {
            T binv = D[i][s]*inv;
            F0R(j,n+2) if (j != s) D[i][j] -= D[r][j]*binv;
            D[i][s] = -binv;
        }
        D[r][s] = 1; F0R(j,n+2) D[r][j] *= inv; // scale r-th row
        swap(B[r],N[s]);
    }
};
```

```
bool simplex(int phase) {
    int x = m+phase-1;
    while (1) { // if phase=1, ignore artificial variable
        int s = -1; F0R(j,n+1) if (N[j] != -phase) ltj(D[x]);
        // find most negative col for nonbasic (NB) variable
        if (D[x][s] >= -eps) return 1;
        // can't get better sol by increasing NB variable
        int r = -1;
        F0R(i,m) {
            if (D[i][s] <= eps) continue;
            if (r == -1 || mp(D[i][n+1] / D[i][s], B[i])
                < mp(D[r][n+1] / D[r][s], B[r])) r = i;
            // find smallest positive ratio
        } // -> max increase in NB variable
        if (r == -1) return 0; // objective is unbounded
        pivot(r,s);
    }
}

T solve(vd& x) { // 1. check if x=0 feasible
    int r = 0; FOR(i,1,m) if (D[i][n+1] < D[r][n+1]) r = i;
    if (D[r][n+1] < -eps) { // if not, find feasible start
        pivot(r,n); // make artificial variable basic
        assert(simplex(2)); // I think this will always be true??
        if (D[m+1][n+1] < -eps) return -inf;
        // D[m+1][n+1] is max possible value of the negation of
        // artificial variable, optimal value should be zero
        // if exists feasible solution
        F0R(i,m) if (B[i] == -1) { // artificial var basic
            int s = 0; FOR(j,1,n+1) ltj(D[i]); // -> nonbasic
            pivot(i,s);
        }
    }
    bool ok = simplex(1); x = vd(n);
    F0R(i,m) if (B[i] < n) x[B[i]] = D[i][n+1];
    return ok ? D[m][n+1] : inf;
}
};
```

Graphs (7)

Erdos-Gallai:  $d_1 \geq \dots \geq d_n$  can be degree sequence of simple graph on n vertices iff their sum is even and  $\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min(d_i, k), \forall 1 \leq k \leq n$ .

7.1 Basics

DSU.h  
**Description:** Disjoint Set Union with path compression and union by size. Add edges and test connectivity. Use for Kruskal’s or Boruvka’s minimum spanning tree.  
**Time:**  $\mathcal{O}(\alpha(N))$

```
struct DSU {
    vi e; void init(int N) { e = vi(N,-1); }
    int get(int x) { return e[x] < 0 ? x : e[x] = get(e[x]); }
    bool sameSet(int a, int b) { return get(a) == get(b); }
    int size(int x) { return -e[get(x)]; }
    bool unite(int x, int y) { // union by size
        x = get(x), y = get(y); if (x == y) return 0;
        if (e[x] > e[y]) swap(x,y);
        e[x] += e[y]; e[y] = x; return 1;
    }
};
```

NegativeCycle.h  
**Description:** use Bellman-Ford (make sure no underflow)

```
688ec8, 11 lines
```

```
vi negCyc(int N, V<pair<pi,int>> ed) {
    vl d(N); vi p(N); int x = -1;
    rep(N) {
        x = -1; each(t,ed) if (ckmin(d[t.f.s],d[t.f.f]+t.s))
            p[t.f.s] = t.f.f, x = t.f.s;
        if (x == -1) return {};
    }
    rep(N) x = p[x]; // enter cycle
    vi cyc{x}; while (p[cyc.bk] != x) cyc.pb(p[cyc.bk]);
    reverse(all(cyc)); return cyc;
}
```

7.2 Trees

LCAjump.h

**Description:** Calculates least common ancestor in tree with verts  $0 \dots N-1$  and root  $R$  using binary jumping.

**Memory:**  $\mathcal{O}(N \log N)$

**Time:**  $\mathcal{O}(N \log N)$  build,  $\mathcal{O}(\log N)$  query

6b0ee9, 28 lines

```
struct LCA {
    int N; V<vi> par, adj; vi depth;
    void init(int _N) { N = _N;
        int d = 1; while ((1<=d) < N) ++d;
        par.assign(d,vi(N)); adj.rsz(N); depth.rsz(N);
    }
    void ae(int x, int y) { adj[x].pb(y), adj[y].pb(x); }
    void gen(int R = 0) { par[0][R] = R; dfs(R); }
    void dfs(int x = 0) {
        FOR(i,1,sz(par)) par[i][x] = par[i-1][par[i-1][x]];
        each(y,adj[x]) if (y != par[0][x])
            depth[y] = depth[par[0][y]=x]+1, dfs(y);
    }
    int jmp(int x, int d) {
        FOR(i,sz(par)) if ((d>>i)&1) x = par[i][x];
        return x; }
    int lca(int x, int y) {
        if (depth[x] < depth[y]) swap(x,y);
        x = jmp(x,depth[x]-depth[y]); if (x == y) return x;
        R0F(i,sz(par)) {
            int X = par[i][x], Y = par[i][y];
            if (X != Y) x = X, y = Y;
        }
        return par[0][x];
    }
    int dist(int x, int y) { // # edges on path
        return depth[x]+depth[y]-2*depth[lca(x,y)]; }
};
```

LCArm q.h

**Description:** Euler Tour LCA. Compress takes a subset  $S$  of nodes and computes the minimal subtree that contains all the nodes pairwise LCAs and compressing edges. Returns a list of (par, orig.index) representing a tree rooted at 0. The root points to itself.

**Time:**  $\mathcal{O}(N \log N)$  build,  $\mathcal{O}(1)$  LCA,  $\mathcal{O}(|S| \log |S|)$  compress

"RMQ.h" e5a035, 28 lines

```
struct LCA {
    int N; V<vi> adj;
    vi depth, pos, par, rev; // rev is for compress
    vpi tmp; RMQ<pi> r;
    void init(int _N) { N = _N; adj.rsz(N);
        depth = pos = par = rev = vi(N); }
    void ae(int x, int y) { adj[x].pb(y), adj[y].pb(x); }
    void dfs(int x) {
        pos[x] = sz(tmp); tmp.pb(depth[x],x);
        each(y,adj[x]) if (y != par[x]) {
            depth[y] = depth[par[y]=x]+1, dfs(y);
            tmp.pb(depth[x],x); }
    }
```

```
void gen(int R = 0) { par[R] = R; dfs(R); r.init(tmp); }
int lca(int u, int v){
    u = pos[u], v = pos[v]; if (u > v) swap(u,v);
    return r.query(u,v).s; }
int dist(int u, int v) {
    return depth[u]+depth[v]-2*depth[lca(u,v)]; }
vpi compress(vi S) {
    auto cmp = [&](int a, int b) { return pos[a] < pos[b]; };
    sort(all(S),cmp); R0F(i,sz(S)-1) S.pb(lca(S[i],S[i+1]));
    sort(all(S),cmp); S.erase(unique(all(S)),end(S));
    vpi ret{{0,S[0]}}; F0R(i,sz(S)) rev[S[i]] = i;
    FOR(i,1,sz(S)) ret.pb(lca(S[i-1],S[i]),S[i]);
    return ret;
}
};
```

HLD.h

**Description:** Heavy-Light Decomposition, add val to verts and query sum in path/subtree.

**Time:** any tree path is split into  $\mathcal{O}(\log N)$  parts

"LazySeg.h" 1802e2, 48 lines

```
template<int SZ, bool VALS_IN_EDGES> struct HLD {
    int N; vi adj[SZ];
    int par[SZ], root[SZ], depth[SZ], sz[SZ], ti;
    int pos[SZ]; vi rpos; // rpos not used but could be useful
    void ae(int x, int y) { adj[x].pb(y), adj[y].pb(x); }
    void dfsSz(int x) {
        sz[x] = 1;
        each(y,adj[x]) {
            par[y] = x; depth[y] = depth[x]+1;
            adj[y].erase(find(all(adj[y]),x));
            dfsSz(y); sz[x] += sz[y];
            if (sz[y] > sz[adj[x][0]]) swap(y,adj[x][0]);
        }
    }
    void dfsHld(int x) {
        pos[x] = ti++; rpos.pb(x);
        each(y,adj[x]) {
            root[y] = (y == adj[x][0] ? root[x] : y);
            dfsHld(y); }
    }
    void init(int _N, int R = 0) { N = _N;
        par[R] = depth[R] = ti = 0; dfsSz(R);
        root[R] = R; dfsHld(R);
    }
    int lca(int x, int y) {
        for (; root[x] != root[y]; y = par[root[y]])
            if (depth[root[x]] > depth[root[y]]) swap(x,y);
        return depth[x] < depth[y] ? x : y;
    }
};
LazySeg<ll,SZ> tree; // segtree for sum
template<class BinaryOp>
void processPath(int x, int y, BinaryOp op) {
    for (; root[x] != root[y]; y = par[root[y]]) {
        if (depth[root[x]] > depth[root[y]]) swap(x,y);
        op(pos[root[y]],pos[y]); }
    if (depth[x] > depth[y]) swap(x,y);
    op(pos[x]+VALS_IN_EDGES,pos[y]);
}
void modifyPath(int x, int y, int v) {
    processPath(x,y,[this,&v](int l, int r) {
        tree.upd(l,r,v); }); }
ll queryPath(int x, int y) {
    ll res = 0; processPath(x,y,[this,&res](int l, int r) {
        res += tree.query(l,r); });
    return res; }
void modifySubtree(int x, int v) {
    tree.upd(pos[x]+VALS_IN_EDGES,pos[x]+sz[x]-1,v); }
};
```

Centroid.h

**Description:** The centroid of a tree of size  $N$  is a vertex such that after removing it, all resulting subtrees have size at most  $\frac{N}{2}$ . Supports updates in the form “add 1 to all verts  $v$  such that  $dist(x,v) \leq y$ .”

**Memory:**  $\mathcal{O}(N \log N)$

**Time:**  $\mathcal{O}(N \log N)$  build,  $\mathcal{O}(\log N)$  update and query

907e21, 54 lines

```
void ad(vi& a, int b) { ckmin(b,sz(a)-1); if (b>=0) a[b]++; }
void prop(vi& a) { R0F(i,sz(a)-1) a[i] += a[i+1]; }
template<int SZ> struct Centroid {
    vi adj[SZ]; void ae(int a,int b){adj[a].pb(b),adj[b].pb(a);}
    bool done[SZ]; // processed as centroid yet
    int N,sub[SZ],cen[SZ],lev[SZ]; // subtree size, centroid anc
    int dist[32-__builtin_clz(SZ)][SZ]; // dists to all ancs
    vi stor[SZ], STOR[SZ];
    void dfs(int x, int p) { sub[x] = 1;
        each(y,adj[x]) if (!done[y] && y != p)
            dfs(y,x), sub[x] += sub[y];
    }
    int centroid(int x) {
        dfs(x,-1);
        for (int sz = sub[x];; ) {
            pi mx = {0,0};
            each(y,adj[x]) if (!done[y] && sub[y] < sub[x])
                ckmax(mx,{sub[y],y});
            if (mx.f*2 <= sz) return x;
            x = mx.s;
        }
    }
    void genDist(int x, int p, int lev) {
        dist[lev][x] = dist[lev][p]+1;
        each(y,adj[x]) if (!done[y] && y != p) genDist(y,x,lev); }
    void gen(int CEN, int _x) { // CEN = centroid above x
        int x = centroid(_x); done[x] = 1; cen[x] = CEN;
        sub[x] = sub[_x]; lev[x] = (CEN == -1 ? 0 : lev[CEN]+1);
        dist[lev[x]][x] = 0;
        stor[x].rsz(sub[x]),STOR[x].rsz(sub[x]+1);
        each(y,adj[x]) if (!done[y]) genDist(y,x,lev[x]);
        each(y,adj[x]) if (!done[y]) gen(x,y);
    }
    void init(int _N) { N = _N; FOR(i,1,N+1) done[i] = 0;
        gen(-1,1); } // start at vert 1
    void upd(int x, int y) {
        int cur = x, pre = -1;
        R0F(i,lev[x]+1) {
            ad(stor[cur],y-dist[i][x]);
            if (pre != -1) ad(STOR[pre],y-dist[i][x]);
            if (i > 0) pre = cur, cur = cen[cur];
        }
    } // call propAll() after all updates
    void propAll() { FOR(i,1,N+1) prop(stor[i]), prop(STOR[i]); }
    int query(int x) { // get value at vertex x
        int cur = x, pre = -1, ans = 0;
        R0F(i,lev[x]+1) { // if pre != -1, subtract those from
            ans += stor[cur][dist[i][x]]; // same subtree
            if (pre != -1) ans -= STOR[pre][dist[i][x]];
            if (i > 0) pre = cur, cur = cen[cur];
        }
        return ans;
    }
};
```

7.2.1 SqrtDecompton

HLD generally suffices. If not, here are some common

strategies to build the tree after every  $\sqrt{N}$  queries.

- Consider vertices with  $>$  or  $< \sqrt{N}$  degree separately.
- For subtree updates, note that there are  $O(\sqrt{N})$  distinct sizes among child subtrees of any node.

**Block Tree:** Use a DFS to split edges into contiguous groups of size  $\sqrt{N}$  to  $2\sqrt{N}$ .

**Mo’s Algorithm for Tree Paths:** Maintain an array of vertices where each one appears twice, once when a DFS enters the vertex (st) and one when the DFS exists (en). For a tree path  $u \leftrightarrow v$  such that  $st[u] < st[v]$ ,

- If  $u$  is an ancestor of  $v$ , query  $[st[u], st[v]]$ .
- Otherwise, query  $[en[u], st[v]]$  and consider  $LCA(u, v)$  separately.

Solutions with worse complexities can be faster if you optimize the operations that are performed most frequently. Use arrays instead of vectors whenever possible. Iterating over an array in order is faster than iterating through the same array in some other order (ex. one given by a random permutation) or DFSing on a tree of the same size. Also, the difference between  $\sqrt{N}$  and the optimal block (or buffer) size can be quite large. Try up to 5x smaller or larger (at least).

7.3 DFS Algorithms

**EulerPath.h**  
**Description:** Eulerian path starting at src if it exists, visits all edges exactly once. Works for both directed and undirected. Returns vector of {vertex,label of edge to vertex}. Second element of first pair is always -1.  
**Time:**  $\mathcal{O}(N + M)$

```
template<bool directed> struct Euler {
    int N; V<vpi> adj; V<vpi::iterator> its; vb used;
    void init(int _N) { N = _N; adj.rsz(N); }
    void ae(int a, int b) {
        int M = sz(used); used.pb(0);
        adj[a].eb(b,M); if (!directed) adj[b].eb(a,M); }
    vpi solve(int src = 0) {
        its.rsz(N); F0R(i,N) its[i] = begin(adj[i]);
        vpi ans, s{{src,-1}}; // {{vert,prev vert},edge label}
        int lst = -1; // ans generated in reverse order
        while (sz(s)) {
            int x = s.bk.f; auto& it=its[x], en=end(adj[x]);
            while (it != en && used[it->s]) ++it;
            if (it == en) { // no more edges out of vertex
                if (lst != -1 && lst != x) return {};
                // not a path, no tour exists
                ans.pb(s.bk); s.pop_back(); if (sz(s)) lst=s.bk.f;
            } else s.pb(*it), used[it->s] = 1;
        } // must use all edges
        if (sz(ans) != sz(used)+1) return {};
        reverse(all(ans)); return ans;
    }
};
```

**SCCT.h**  
**Description:** Tarjan’s, DFS once to generate strongly connected components in topological order.  $a, b$  in same component if both  $a \rightarrow b$  and  $b \rightarrow a$  exist. Uses less memory than Kosaraju b/c doesn’t store reverse edges.  
**Time:**  $\mathcal{O}(N + M)$

```
struct SCC {
    int N, ti = 0; V<vi> adj;
    vi disc, comp, stk, comps;
    void init(int _N) { N = _N; adj.rsz(N);
        disc.rsz(N), comp.rsz(N,-1); }
    void ae(int x, int y) { adj[x].pb(y); }
    int dfs(int x) {
        int low = disc[x] = ++ti; stk.pb(x);
        each(y,adj[x]) if (comp[y] == -1) // comp[y] == -1,
            ckmin(low,disc[y]?:dfs(y)); // disc[y] != 0 -> in stack
        if (low == disc[x]) { // make new SCC
            // pop off stack until you find x
            comps.pb(x); for (int y = -1; y != x;)
                comp[y = stk.bk] = x, stk.pop_back();
        }
        return low;
    }
    void gen() {
        F0R(i,N) if (!disc[i]) dfs(i);
        reverse(all(comps));
    }
};
```

**TwoSAT.h**  
**Description:** Calculates a valid assignment to boolean variables  $a, b, c, \dots$  to a 2-SAT problem, so that an expression of the type  $(a||b)&&(!a||c)&&(d||!b)&& \dots$  becomes true, or reports that it is unsatisfiable. Negated variables are represented by bit-inversions ( $\sim x$ ).  
**Usage:** TwoSat ts;  
ts.either(0, ~3); // Var 0 is true or var 3 is false  
ts.setVal(2); // Var 2 is true  
ts.atMostOne({0,~1,2}); //  $\leq 1$  of vars 0, ~1 and 2 are true  
ts.solve(N); // Returns true iff it is solvable  
ts.ans[0..N-1] holds the assigned values to the vars  
"scc.h" f5f0f3d, 31 lines

```
struct TwoSAT {
    int N = 0; vpi edges;
    void init(int _N) { N = _N; }
    int addVar() { return N++; }
    void either(int x, int y) {
        x = max(2*x,-1-2*x), y = max(2*y,-1-2*y);
        edges.eb(x,y); }
    void implies(int x, int y) { either(~x,y); }
    void must(int x) { either(x,x); }
    void atMostOne(const vi& li) {
        if (sz(li) <= 1) return;
        int cur = ~li[0];
        F0R(i,2,sz(li)) {
            int next = addVar();
            either(cur,~li[i]); either(cur,next);
            either(~li[i],next); cur = ~next;
        }
        either(cur,~li[1]);
    }
    vb solve() {
        SCC S; S.init(2*N);
        each(e,edges) S.ae(e.f^1,e.s), S.ae(e.s^1,e.f);
        S.gen(); reverse(all(S.comps)); // reverse topo order
        for (int i = 0; i < 2*N; i += 2)
            if (S.comp[i] == S.comp[i^1]) return {};
        vi tmp(2*N); each(i,S.comps) if (!tmp[i])
            tmp[i] = 1, tmp[S.comp[i^1]] = -1;
        vb ans(N); F0R(i,N) ans[i] = tmp[S.comp[2*i]] == 1;
        return ans;
    }
};
```

**BCC.h**  
**Description:** Biconnected components of edges. Removing any vertex in BCC doesn’t disconnect it. To get block-cut tree, create a bipartite graph with the original vertices on the left and a vertex for each BCC on the right. Draw edge  $u \leftrightarrow v$  if  $u$  is contained within the BCC for  $v$ . Self-loops are not included in any BCC while BCCs of size 1 represent bridges.  
**Time:**  $\mathcal{O}(N + M)$

```
0625a6, 35 lines
struct BCC {
    V<vpi> adj; vpi ed;
    V<vi> edgeSets, vertSets; // edges for each bcc
    int N, ti = 0; vi disc, stk;
    void init(int _N) { N = _N; disc.rsz(N), adj.rsz(N); }
    void ae(int x, int y) {
        adj[x].eb(y,sz(ed)), adj[y].eb(x,sz(ed)), ed.eb(x,y); }
    int dfs(int x, int p = -1) { // return lowest disc
        int low = disc[x] = ++ti;
        each(e,adj[x]) if (e.s != p) {
            if (!disc[e.f]) {
                stk.pb(e.s); // disc[x] < LOW -> bridge
                int LOW = dfs(e.f,e.s); ckmin(low,LOW);
                if (disc[x] <= LOW) { // get edges in bcc
                    edgeSets.eb(); vi& tmp = edgeSets.bk; // new bcc
                    for (int y = -1; y != e.s; )
                        tmp.pb(y = stk.bk), stk.pop_back();
                }
            } else if (disc[e.f] < disc[x]) // back-edge
                ckmin(low,disc[e.f]), stk.pb(e.s);
        }
        return low;
    }
    void gen() {
        F0R(i,N) if (!disc[i]) dfs(i);
        vb in(N);
        each(c,edgeSets) { // edges contained within each BCC
            vertSets.eb(); // so you can easily create block cut tree
            auto ad = [&](int x) {
                if (!in[x]) in[x] = 1, vertSets.bk.pb(x); };
            each(e,c) ad(ed[e].f), ad(ed[e].s);
            each(e,c) in[ed[e].f] = in[ed[e].s] = 0;
        }
    }
};
```

**MaximalCliques.h**  
**Description:** Used only once. Finds all maximal cliques.  
**Time:**  $\mathcal{O}(3^{N/3})$

```
f5cd93, 16 lines
using B = bitset<128>; B adj[128];
int N;
// possibly in clique, not in clique, in clique
void cliques(B P = ~B(), B X={}, B R={}) {
    if (!P.any()) {
        if (!X.any()) // do smth with R
            return;
    }
    int q = (P|X)._Find_first();
    // clique must contain q or non-neighbor of q
    B cands = P&~adj[q];
    F0R(i,N) if (cands[i]) {
        R[i] = 1; cliques(P&adj[i],X&adj[i],R);
        R[i] = P[i] = 0; X[i] = 1;
    }
}
```

7.4 Flows

**Konig’s Theorem:** In a bipartite graph, max matching = min vertex cover.

**Dilworth’s Theorem:** For any partially ordered set, the sizes of the max antichain and of the min chain decomposition are equal. Equivalent to Konig’s theorem on the bipartite graph  $(U, V, E)$  where  $U = V = S$  and  $(u, v)$  is an edge when  $u < v$ . Those vertices outside the min vertex cover in both  $U$  and  $V$  form a max antichain.

Dinic.h  
**Description:** Fast flow. After computing flow, edges  $\{u, v\}$  such that  $lev[u] \neq -1$ ,  $lev[v] = -1$  are part of min cut. Use reset and rcap for Gomory-Hu.  
**Time:**  $\mathcal{O}(N^2M)$  flow,  $\mathcal{O}(M\sqrt{N})$  bipartite matching

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <pre>struct Dinic {     using F = ll; // flow type     struct Edge { int to; F flo, cap; };     int N; V&lt;Edge&gt; eds; V&lt;vi&gt; adj;     void init(int _N) { N = _N; adj.rsz(N), cur.rsz(N); }     void ae(int u, int v, F cap, F rcap = 0) { assert(min(cap,         ↪rcap) &gt;= 0);         adj[u].pb(sz(eds)); eds.pb({v,0,cap});         adj[v].pb(sz(eds)); eds.pb({u,0,rcap});     }     vi lev; V&lt;vi::iterator&gt; cur;     bool bfs(int s, int t) { // level = shortest distance from         ↪source         lev = vi(N,-1); F0R(i,N) cur[i] = begin(adj[i]);         queue&lt;int&gt; q({s}); lev[s] = 0;         while (sz(q)) { int u = q.ft; q.pop();             each(e,adj[u]) { const Edge&amp; E = eds[e];                 int v = E.to; if (lev[v] &lt; 0 &amp;&amp; E.flo &lt; E.cap)                     q.push(v), lev[v] = lev[u]+1;             }         }         return lev[t] &gt;= 0;     }     F dfs(int v, int t, F flo) {         if (v == t) return flo;         for (; cur[v] != end(adj[v]); cur[v]++) {             Edge&amp; E = eds[*cur[v]];             if (lev[E.to]!=lev[v]+1  E.flo==E.cap) continue;             F df = dfs(E.to,t,min(flo,E.cap-E.flo));             if (df) { E.flo += df; eds[*cur[v]^1].flo -= df;                 return df; } // saturated &gt;=1 one edge         }         return 0;     }     F maxFlow(int s, int t) {         F tot = 0; while (bfs(s,t)) while (F df =             dfs(s,t,numeric_limits&lt;F&gt;::max())) tot += df;         return tot;     } };</pre> | b7b370, 38 lines |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|

|                                                                                                                                                                                                                                                                                                                       |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <pre>each(p,ed_locs) { // reset capacities     auto&amp; e = D.adj.at(p.f).at(p.s);     auto&amp; e_rev = D.adj.at(e.to).at(e.rev);     e.cap = e_rev.cap = (e.cap+e_rev.cap)/2; } ans.pb({{i,par[i]},D.maxFlow(i,par[i])}); FOR(j,i+1,N) if (par[j] == par[i] &amp;&amp; D.lev[j]) par[j] = i; } return ans; }</pre> |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

MCMF.h  
**Description:** Minimum-cost maximum flow, assumes no negative cycles. It is possible to choose negative edge costs such that the first run of Dijkstra is slow, but this hasn’t been an issue in the past. Edge weights  $\geq 0$  for every subsequent run. To get flow through original edges, assign ID’s during ae.  
**Time:** Ignoring first run of Dijkstra,  $\mathcal{O}(FM \log M)$  if caps are integers and  $F$  is max flow.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <pre>struct MCMF {     using F = ll; using C = ll; // flow type, cost type     struct Edge { int to; F flo, cap; C cost; };     int N; V&lt;C&gt; p, dist; vi pre; V&lt;Edge&gt; eds; V&lt;vi&gt; adj;     void init(int _N) { N = _N;         p.rsz(N), dist.rsz(N), pre.rsz(N), adj.rsz(N); }     void ae(int u, int v, F cap, C cost) { assert(cap &gt;= 0);         adj[u].pb(sz(eds)); eds.pb({v,0,cap,cost});         adj[v].pb(sz(eds)); eds.pb({u,0,0,-cost});     } // use asserts, don’t try smth dumb     bool path(int s, int t) { // find lowest cost path to send         ↪flow through         const C inf = numeric_limits&lt;C&gt;::max(); F0R(i,N) dist[i] =             ↪inf;         using T = pair&lt;C,int&gt;; priority_queue&lt;T,vector&lt;T&gt;,greater&lt;T             ↪&gt;&gt;&gt; todo;         todo.push({dist[s] = 0,s});         while (sz(todo)) { // Dijkstra             T x = todo.top(); todo.pop(); if (x.f &gt; dist[x.s])                 ↪continue;             each(e,adj[x.s]) { const Edge&amp; E = eds[e]; // all weights                 ↪should be non-negative                 if (E.flo &lt; E.cap &amp;&amp; ckmin(dist[E.to],x.f+E.cost+p[x.s                     ↪]-p[E.to]))                     pre[E.to] = e, todo.push({dist[E.to],E.to});             }         } // if costs are doubles, add some EPS so you         // don’t traverse ~0-weight cycle repeatedly         return dist[t] != inf; // return flow     }     pair&lt;F,C&gt; calc(int s, int t) { assert(s != t);         F0R(_,N) F0R(e,sz(eds)) { const Edge&amp; E = eds[e]; //             ↪Bellman-Ford             if (E.cap) ckmin(p[E.to],p[eds[e^1].to]+E.cost); }         F totFlow = 0; C totCost = 0;         while (path(s,t)) { // p -&gt; potentials for Dijkstra             F0R(i,N) p[i] += dist[i]; // don’t matter for unreachable                 ↪nodes             F df = numeric_limits&lt;F&gt;::max();             for (int x = t; x != s; x = eds[pre[x]^1].to) {                 const Edge&amp; E = eds[pre[x]]; ckmin(df,E.cap-E.flo); }             totFlow += df; totCost += (p[t]-p[s])*df;             for (int x = t; x != s; x = eds[pre[x]^1].to)                 eds[pre[x]].flo += df, eds[pre[x]^1].flo -= df;         } // get max flow you can send along path         return {totFlow,totCost};     } };</pre> | 072c1b, 40 lines |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|

## 7.5 Matching

Hungarian.h  
**Description:** Given J jobs and W workers (J ≤ W), computes the minimum cost to assign each prefix of jobs to distinct workers.  
@tparam T a type large enough to represent integers on the order of J \* max(—C—) @param C a matrix of dimensions JxW such that C[j][w] = cost to assign j-th job to w-th worker (possibly negative)  
@return a vector of length J, with the j-th entry equaling the minimum cost to assign the first (j+1) jobs to distinct workers  
**Time:**  $\mathcal{O}(J^2W)$

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <pre>template &lt;class T&gt; vector&lt;T&gt; hungarian(const vector&lt;vector&lt;T&gt;&gt;     ↪&amp;C) {     const int J = (int)size(C), W = (int)size(C[0]);     assert(J &lt;= W);     vector&lt;int&gt; job(W + 1, -1);     vector&lt;T&gt; ys(J), yt(W + 1);     vector&lt;T&gt; answers;     const T inf = numeric_limits&lt;T&gt;::max();     for (int j_cur = 0; j_cur &lt; J; ++j_cur) {         int w_cur = W;         job[w_cur] = j_cur;         vector&lt;T&gt; min_to(W + 1, inf);         vector&lt;int&gt; prv(W + 1, -1);         vector&lt;bool&gt; in_Z(W + 1);         while (job[w_cur] != -1) {             in_Z[w_cur] = true;             const int j = job[w_cur];             T delta = inf;             int w_next;             for (int w = 0; w &lt; W; ++w) {                 if (!in_Z[w]) {                     if (ckmin(min_to[w], C[j][w] - ys[j] - yt[w]))                         prv[w] = w_cur;                     if (ckmin(delta, min_to[w])) w_next = w;                 }             }             for (int w = 0; w &lt;= W; ++w) {                 if (in_Z[w]) ys[job[w]] += delta, yt[w] -= delta;                 else min_to[w] -= delta;             }             w_cur = w_next;         }         for (int w; w_cur != -1; w_cur = w) job[w_cur] = job[w =             ↪prv[w_cur]];         answers.push_back(-yt[W]);     }     return answers; }</pre> | f382f7, 36 lines |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|

GeneralMatchBlossom.h  
**Description:** Variant on Gabow’s Impl of Edmond’s Blossom Algorithm. General unweighted max matching with 1-based indexing. If white[v] = 0 after solve() returns, v is part of every max matching.  
**Time:**  $\mathcal{O}(NM)$ , faster in practice

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <pre>struct MaxMatching {     int N; V&lt;vi&gt; adj;     V&lt;int&gt; mate, first; vb white; vpi label;     void init(int _N) { N = _N; adj = V&lt;vi&gt;(N+1);         mate = first = vi(N+1); label = vpi(N+1); white = vb(N+1);         ↪;     void ae(int u, int v) { adj.at(u).pb(v), adj.at(v).pb(u); }     int group(int x) { if (white[first[x]]) first[x] = group(         ↪first[x]);         return first[x]; }     void match(int p, int b) {         swap(b,mate[p]); if (mate[b] != p) return;</pre> | fd5cc7, 50 lines |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|

```
if (!label[p].s) mate[b] = label[p].f, match(label[p].f,b);
↪ // vertex label
else match(label[p].f,label[p].s), match(label[p].s,label[p]
↪).f); // edge label
}
bool augment(int st) { assert(st);
white[st] = 1; first[st] = 0; label[st] = {0,0};
queue<int> q; q.push(st);
while (!q.empty()) {
int a = q.ft; q.pop(); // outer vertex
each(b,adj[a]) { assert(b);
if (white[b]) { // two outer vertices, form blossom
int x = group(a), y = group(b), lca = 0;
while (x||y) {
if (y) swap(x,y);
if (label[x] == pi{a,b}) { lca = x; break; }
label[x] = {a,b}; x = group(label[mate[x]].first);
}
for (int v: {group(a),group(b)}) while (v != lca) {
assert(!white[v]); // make everything along path
↪white
q.push(v); white[v] = true; first[v] = lca;
v = group(label[mate[v]].first);
}
} else if (!mate[b]) { // found augmenting path
mate[b] = a; match(a,b); white = vb(N+1); // reset
return true;
} else if (!white[mate[b]]) {
white[mate[b]] = true; first[mate[b]] = b;
label[b] = {0,0}; label[mate[b]] = pi{a,0};
q.push(mate[b]);
}
}
}
return false;
}
int solve() {
int ans = 0;
FOR(st,1,N+1) if (!mate[st]) ans += augment(st);
FOR(st,1,N+1) if (!mate[st] && !white[st]) assert(!augment(
↪st));
return ans;
}
};
```

GeneralWeightedMatch.h

**Description:** General max weight max matching with 1-based indexing. Edge weights must be positive, combo of UnweightedMatch and Hungarian. **Time:**  $O(N^3)$ ?

120873, 145 lines

```
template<int SZ> struct WeightedMatch {
struct edge { int u,v,w; }; edge g[SZ*2][SZ*2];
void ae(int u, int v, int w) { g[u][v].w = g[v][u].w = w; }
int N,NX,lab[SZ*2],match[SZ*2],slack[SZ*2],st[SZ*2];
int par[SZ*2],floFrom[SZ*2][SZ],S[SZ*2],aux[SZ*2];
vi flo[SZ*2]; queue<int> q;
void init(int _N) { N = _N; // init all edges
FOR(u,1,N+1) FOR(v,1,N+1) g[u][v] = {u,v,0}; }
int eDelta(edge e) { // >= 0 at all times
return lab[e.u]+lab[e.v]-g[e.u][e.v].w*2; }
void updSlack(int u, int x) { // smallest edge -> blossom x
if (!slack[x] || eDelta(g[u][x]) < eDelta(g[slack[x]][x]))
slack[x] = u; }
void setSlack(int x) {
slack[x] = 0; FOR(u,1,N+1) if (g[u][x].w > 0
&& st[u] != x && S[st[u]] == 0) updSlack(u,x); }
void qPush(int x) {
if (x <= N) q.push(x);
else each(t,flo[x]) qPush(t); }
```

GeneralWeightedMatch MaxMatchFast

```
void setSt(int x, int b) {
st[x] = b; if (x > N) each(t,flo[x]) setSt(t,b); }
int getPr(int b, int xr) { // get even position of xr
int pr = find(all(flo[b]),xr)-begin(flo[b]);
if (pr&1) { reverse(l+all(flo[b])); return sz(flo[b])-pr; }
return pr; }
void setMatch(int u, int v) { // rearrange flo[u], matches
edge e = g[u][v]; match[u] = e.v; if (u <= N) return;
int xr = floFrom[u][e.u], pr = getPr(u,xr);
F0R(i,pr) setMatch(flo[u][i],flo[u][i^1]);
setMatch(xr,v); rotate(begin(flo[u]),pr+all(flo[u])); }
void augment(int u, int v) { // set matches including u->v
while (1) { // and previous ones
int xnv = st[match[u]]; setMatch(u,v);
if (!xnv) return;
setMatch(xnv,st[par[xnv]]);
u = st[par[xnv]], v = xnv;
}
}
int lca(int u, int v) { // same as in unweighted
static int t = 0; // except maybe return 0
for (++t;u||v;swap(u,v)) {
if (!u) continue;
if (aux[u] == t) return u;
aux[u] = t; u = st[match[u]];
if (u) u = st[par[u]];
}
return 0;
}
void addBlossom(int u, int anc, int v) {
int b = N+1; while (b <= NX && st[b]) ++b;
if (b > NX) ++NX; // new blossom
lab[b] = S[b] = 0; match[b] = match[anc]; flo[b] = {anc};
auto blossom = [&](int x) {
for (int y; x != anc; x = st[par[y]])
flo[b].pb(x), flo[b].pb(y = st[match[x]]), qPush(y);
};
blossom(u); reverse(l+all(flo[b])); blossom(v); setSt(b,b);
// identify all nodes in current blossom
FOR(x,1,NX+1) g[b][x].w = g[x][b].w = 0;
FOR(x,1,N+1) floFrom[b][x] = 0;
each(xs,flo[b]) { // find tightest constraints
FOR(x,1,NX+1) if (g[b][x].w == 0 || eDelta(g[xs][x]) <
eDelta(g[b][x])) g[b][x]=g[xs][x], g[x][b]=g[x][xs];
FOR(x,1,N+1) if (floFrom[xs][x]) floFrom[b][x] = xs;
} // floFrom to deconstruct blossom
setSlack(b); // since didn't qPush everything
}
void expandBlossom(int b) {
each(t,flo[b]) setSt(t,t); // undo setSt(b,b)
int xr = floFrom[b][g[b][par[b]].u], pr = getPr(b,xr);
for(int i = 0; i < pr; i += 2) {
int xs = flo[b][i], xns = flo[b][i+1];
par[xs] = g[xns][xs].u; S[xs] = 1; // no setSlack(xns)?
S[xns] = slack[xs] = slack[xns] = 0; qPush(xns);
}
S[xr] = 1, par[xr] = par[b];
FOR(i,pr+1,sz(flo[b])) { // matches don't change
int xs = flo[b][i]; S[xs] = -1, setSlack(xs); }
st[b] = 0; // blossom killed
}
bool onFoundEdge(edge e) {
int u = st[e.u], v = st[e.v];
if (S[v] == -1) { // v unvisited, matched with smth else
par[v] = e.u, S[v] = 1; slack[v] = 0;
int nu = st[match[v]]; S[nu] = slack[nu] = 0; qPush(nu);
} else if (S[v] == 0) {
int anc = lca(u,v); // if 0 then match found!
if (!anc) return augment(u,v),augment(v,u),1;
}
```

```
addBlossom(u,anc,v);
}
return 0;
}
bool matching() {
q = queue<int>();
FOR(x,1,NX+1) {
S[x] = -1, slack[x] = 0; // all initially unvisited
if (st[x] == x && !match[x]) par[x] = S[x] = 0, qPush(x);
}
if (!sz(q)) return 0;
while (1) {
while (sz(q)) { // unweighted matching with tight edges
int u = q.ft; q.pop(); if (S[st[u]] == 1) continue;
FOR(v,1,N+1) if (g[u][v].w > 0 && st[u] != st[v]) {
if (eDelta(g[u][v]) == 0) { // condition is strict
if (onFoundEdge(g[u][v])) return 1;
} else updSlack(u,st[v]);
}
}
int d = INT_MAX;
FOR(b,N+1,NX+1) if (st[b] == b && S[b] == 1)
ckmin(d,lab[b]/2); // decrease lab[b]
FOR(x,1,NX+1) if (st[x] == x && slack[x]) {
if (S[x] == -1) ckmin(d,eDelta(g[slack[x]][x]));
else if (S[x] == 0) ckmin(d,eDelta(g[slack[x]][x])/2);
} // edge weights shouldn't go below 0
FOR(u,1,N+1) {
if (S[st[u]] == 0) {
if (lab[u] <= d) return 0; // why?
lab[u] -= d;
} else if (S[st[u]] == 1) lab[u] += d;
} // lab has opposite meaning for verts and blossoms
FOR(b,N+1,NX+1) if (st[b] == b && S[b] != -1)
lab[b] += (S[b] == 0 ? 1 : -1)*d*2;
q = queue<int>();
FOR(x,1,NX+1) if (st[x]==x && slack[x] // new tight edge
&& st[slack[x]] != x && eDelta(g[slack[x]][x]) == 0)
if (onFoundEdge(g[slack[x]][x])) return 1;
FOR(b,N+1,NX+1) if (st[b]==b && S[b]==1 && lab[b]==0)
expandBlossom(b); // odd dist blossom taken apart
}
return 0;
}
pair<ll,int> calc() {
NX = N; st[0] = 0; FOR(i,1,2*N+1) aux[i] = 0;
FOR(i,1,N+1) match[i] = 0, st[i] = i, flo[i].clear();
int wMax = 0;
FOR(u,1,N+1) FOR(v,1,N+1)
floFrom[u][v] = (u == v ? u : 0), ckmax(wMax,g[u][v].w);
FOR(u,1,N+1) lab[u] = wMax; // start high and decrease
int num = 0; ll wei = 0; while (matching()) ++num;
FOR(u,1,N+1) if (match[u] && match[u] < u)
wei += g[u][match[u]].w; // edges in matching
return {wei,num};
}
};
```

MaxMatchFast.h

**Description:** Fast bipartite matching.

**Time:**  $O(M\sqrt{N})$

ec6c96, 31 lines

```
vi maxMatch(int L, int R, const vpi& edges) {
V<vi> adj = V<vi>(L);
vi nxt(L,-1), prv(R,-1), lev, ptr;
F0R(i,sz(edges)) adj.at(edges[i].f).pb(edges[i].s);
while (true) {
lev = ptr = vi(L); int max_lev = 0;
queue<int> q; F0R(i,L) if (nxt[i]==-1) lev[i]=1, q.push(i);
```

```
while (sz(q)) {
    int x = q.ft; q.pop();
    for (int y: adj[x]) {
        int z = prv[y];
        if (z == -1) max_lev = lev[x];
        else if (!lev[z]) lev[z] = lev[x]+1, q.push(z);
    }
    if (max_lev) break;
}
if (!max_lev) break;
F0R(i,L) if (lev[i] > max_lev) lev[i] = 0;
auto dfs = [&](auto self, int x) -> bool {
    for (;ptr[x] < sz(adj[x]);++ptr[x]) {
        int y = adj[x][ptr[x]], z = prv[y];
        if (z == -1 || (lev[z] == lev[x]+1 && self(self,z)))
            return nxt[x]=y, prv[y]=x, ptr[x]=sz(adj[x]), 1;
    }
    return 0;
};
F0R(i,L) if (nxt[i] == -1) dfs(dfs,i);
}
vpi ans; F0R(i,L) if (nxt[i] != -1) ans.pb({i,nxt[i]});
return ans;
}
```

7.6 Advanced

ChordalGraphRecognition.h

**Description:** Recognizes graph where every induced cycle has length exactly 3 using maximum adjacency search.

6cc97d, 58 lines

```
int N,M;
set<int> adj[MX];
int cnt[MX];
vi ord, rord;

vi find_path(int x, int y, int z) {
    vi pre(N,-1);
    queue<int> q; q.push(x);
    while (sz(q)) {
        int t = q.ft; q.pop();
        if (adj[t].count(y)) {
            pre[y] = t; vi path = {y};
            while (path.bk != x) path.pb(pre[path.bk]);
            path.pb(z);
            return path;
        }
        each(u,adj[t]) if (u != z && !adj[u].count(z) && pre[u] == -1) {
            pre[u] = t;
            q.push(u);
        }
    }
    assert(0);
}

int main() {
    setIO(); re(N,M);
    F0R(i,M) {
        int a,b; re(a,b);
        adj[a].insert(b), adj[b].insert(a);
    }
    rord = vi(N,-1);
    priority_queue<pi> pq;
    F0R(i,N) pq.push({0,i});
    while (sz(pq)) {
        pi p = pq.top(); pq.pop();
        if (rord[p.s] != -1) continue;
        rord[p.s] = sz(ord); ord.pb(p.s);
        each(t,adj[p.s]) pq.push({++cnt[t],t});
    }
}
```

```
}
assert(sz(ord) == N);
each(z,ord) {
    pi big = {-1,-1};
    each(y,adj[z]) if (rord[y] < rord[z])
        ckmax(big,mp(rord[y],y));
    if (big.f == -1) continue;
    int y = big.s;
    each(x,adj[z]) if (rord[x] < rord[y]) if (!adj[y].count(x))
        -> {
            ps("NO");
            vi v = find_path(x,y,z);
            ps(sz(v));
            each(t,v) pr(t,' ');
            exit(0);
        }
    }
    ps("YES");
    reverse(all(ord));
    each(z,ord) pr(z,' ');
}
```

DominatorTree.h

**Description:** Used only a few times. Assuming that all nodes are reachable from *root*, *a* dominates *b* iff every path from *root* to *b* passes through *a*.

**Time:**  $\mathcal{O}(M \log N)$

4b8836, 41 lines

```
template<int SZ> struct Dominator {
    vi adj[SZ], ans[SZ]; // input edges, edges of dominator tree
    vi radj[SZ], child[SZ], sdomChild[SZ];
    int label[SZ], rlabel[SZ], sdom[SZ], dom[SZ], co = 0;
    int par[SZ], bes[SZ];
    void ae(int a, int b) { adj[a].pb(b); }
    int get(int x) { // DSU with path compression
        // get vertex with smallest sdom on path to root
        if (par[x] != x) {
            int t = get(par[x]); par[x] = par[par[x]];
            if (sdom[t] < sdom[bes[x]]) bes[x] = t;
        }
        return bes[x];
    }
    void dfs(int x) { // create DFS tree
        label[x] = ++co; rlabel[co] = x;
        sdom[co] = par[co] = bes[co] = co;
        each(y,adj[x]) {
            if (!label[y]) {
                dfs(y); child[label[x]].pb(label[y]);
                radj[label[y]].pb(label[x]);
            }
        }
    }
    void init(int root) {
        dfs(root);
        ROF(i,1,co+1) {
            each(j,radj[i]) ckmin(sdom[i],sdom[get(j)]);
            if (i > 1) sdomChild[sdom[i]].pb(i);
            each(j,sdomChild[i]) {
                int k = get(j);
                if (sdom[j] == sdom[k]) dom[j] = sdom[j];
                else dom[j] = k;
            }
            each(j,child[i]) par[j] = i;
        }
        FOR(i,2,co+1) {
            if (dom[i] != sdom[i]) dom[i] = dom[dom[i]];
            ans[rlabel[dom[i]]].pb(rlabel[i]);
        }
    }
};
```

EdgeColor.h

**Description:** Used only once. Naive implementation of Misra & Gries edge coloring. By Vizing's Theorem, a simple graph with max degree *d* can be edge colored with at most *d* + 1 colors

**Time:**  $\mathcal{O}(N^2M)$ , faster in practice

cc2b29, 40 lines

```
template<int SZ> struct EdgeColor {
    int N = 0, maxDeg = 0, adj[SZ][SZ], deg[SZ];
    void init(int _N) { N = _N;
        F0R(i,N) { deg[i] = 0; F0R(j,N) adj[i][j] = 0; } }
    void ae(int a, int b, int c) {
        adj[a][b] = adj[b][a] = c; }
    int delEdge(int a, int b) {
        int c = adj[a][b]; adj[a][b] = adj[b][a] = 0;
        return c; }
    V<bool> genCol(int x) {
        V<bool> col(N+1); F0R(i,N) col[adj[x][i]] = 1;
        return col; }
    int freeCol(int u) {
        auto col = genCol(u); int x = 1;
        while (col[x]) ++x; return x; }
    void invert(int x, int d, int c) {
        F0R(i,N) if (adj[x][i] == d)
            delEdge(x,i), invert(i,c,d), ae(x,i,c); }
    void ae(int u, int v) {
        // check if you can add edge w/o doing any work
        assert(N); ckmax(maxDeg,max(++deg[u],++deg[v]));
        auto a = genCol(u), b = genCol(v);
        FOR(i,1,maxDeg+2) if (!a[i] && !b[i])
            return ae(u,v,i);
        V<bool> use(N); vi fan = {v}; use[v] = 1;
        while (1) {
            auto col = genCol(fan.bk);
            if (sz(fan) > 1) col[adj[fan.bk][u]] = 0;
            int i=0; while (i<N && (use[i] || col[adj[u][i]])) i++;
            if (i < N) fan.pb(i), use[i] = 1;
            else break;
        }
        int c = freeCol(u), d = freeCol(fan.bk); invert(u,d,c);
        int i = 0; while (i < sz(fan) && genCol(fan[i])[d]
            && adj[u][fan[i]] != d) i++;
        assert (i != sz(fan));
        F0R(j,i) ae(u,fan[j],delEdge(u,fan[j+1]));
        ae(u,fan[i],d);
    }
};
```

DirectedMST.h

**Description:** Chu-Liu-Edmonds algorithm. Computes minimum weight directed spanning tree rooted at *r*, edge from *par*[*i*] → *i* for all *i* ≠ *r*. Use DSU with rollback if need to return edges.

**Time:**  $\mathcal{O}(M \log M)$

"DSUrb.h" 5d5c10, 61 lines

```
struct Edge { int a, b; ll w; };
struct Node { // lazy skew heap node
    Edge key; Node *l, *r; ll delta;
    void prop() {
        key.w += delta;
        if (l) l->delta += delta;
        if (r) r->delta += delta;
        delta = 0;
    }
    Edge top() { prop(); return key; }
};
Node *merge(Node *a, Node *b) {
    if (!a || !b) return a ?: b;
    a->prop(), b->prop();
    if (a->key.w > b->key.w) swap(a, b);
    swap(a->l, a->r = merge(b, a->r));
}
```

```
    return a;
}
void pop(Node*& a) { a->prop(); a = merge(a->l, a->r); }
```

```
pair<ll,vi> dmst(int n, int r, const vector<Edge>& g) {
    DSUrb dsu; dsu.init(n);
    vector<Node*> heap(n); // store edges entering each vertex
    // in increasing order of weight
    each(e,g) heap[e.b] = merge(heap[e.b], new Node(e));
    ll res = 0; vi seen(n,-1); seen[r] = r;
    vpi in(n,{-1,-1}); // edge entering each vertex in MST
    vector<pair<int,vector<Edge>>> cycs;
    FOR(s,n) {
        int u = s, w;
        vector<pair<int,Edge>> path;
        while (seen[u] < 0) {
            if (!heap[u]) return {-1,{};};
            seen[u] = s;
            Edge e = heap[u]->top(); path.pb({u,e});
            heap[u]->delta -= e.w, pop(heap[u]);
            res += e.w, u = dsu.get(e.a);
            if (seen[u] == s) { // found cycle, contract
                Node* cyc = 0; cycs.eb();
                do {
                    cyc = merge(cyc, heap[w = path.bk.f]);
                    cycs.bk.s.pb(path.bk.s);
                    path.pop_back();
                } while (dsu.unite(u,w));
                u = dsu.get(u); heap[u] = cyc, seen[u] = -1;
                cycs.bk.f = u;
            }
            each(t,path) in[dsu.get(t.s.b)] = {t.s.a,t.s.b};
        } // found path from root to s, done
        while (sz(cycs)) { // expand cycs to restore sol
            auto c = cycs.bk; cycs.pop_back();
            pi inEdge = in[c.f];
            each(t,c.s) dsu.rollback();
            each(t,c.s) in[dsu.get(t.b)] = {t.a,t.b};
            in[dsu.get(inEdge.s)] = inEdge;
        }
        vi par(n); FOR(i,n) par[i] = in[i].f;
        // i == r ? in[i].s == -1 : in[i].s == i
        return {res,par};
    }
```

**LCT.h**  
**Description:** Link-Cut Tree. Given a function  $f(1 \dots N) \rightarrow 1 \dots N$ , evaluates  $f^b(a)$  for any  $a, b$ . sz is for path queries; sub, vsub are for subtree queries. x->access() brings x to the top and propagates it; its left subtree will be the path from x to the root and its right subtree will be empty. Then sub will be the number of nodes in the connected component of x and vsub will be the number of nodes under x. Use makeRoot for arbitrary path queries.  
**Usage:** FOR(i,1,N+1)LCT[i]=new snode(i); link(LCT[1],LCT[2],1);  
**Time:**  $\mathcal{O}(\log N)$

e24bf7, 115 lines

```
typedef struct snode* sn;
struct snode { ////////// VARIABLES
    sn p, c[2]; // parent, children
    sn extra; // extra cycle node for "The Applicant"
    bool flip = 0; // subtree flipped or not
    int val, sz; // value in node, # nodes in current splay tree
    int sub, vsub = 0; // vsub stores sum of virtual children
    snode(int _val) : val(_val) {
        p = c[0] = c[1] = extra = NULL; calc(); }
    friend int getSz(sn x) { return x?x->sz:0; }
    friend int getSub(sn x) { return x?x->sub:0; }
    void prop() { // lazy prop
```

```
    if (!flip) return;
    swap(c[0],c[1]); flip = 0;
    FOR(i,2) if (c[i]) c[i]->flip ^= 1;
}
void calc() { // recalc vals
    FOR(i,2) if (c[i]) c[i]->prop();
    sz = 1+getSz(c[0])+getSz(c[1]);
    sub = 1+getSub(c[0])+getSub(c[1])+vsub;
}
////////// SPLAY TREE OPERATIONS
int dir() {
    if (!p) return -2;
    FOR(i,2) if (p->c[i] == this) return i;
    return -1; // p is path-parent pointer
} // -> not in current splay tree
// test if root of current splay tree
bool isRoot() { return dir() < 0; }
friend void setLink(sn x, sn y, int d) {
    if (y) y->p = x;
    if (d >= 0) x->c[d] = y; }
void rot() { // assume p and p->p propagated
    assert(!isRoot()); int x = dir(); sn pa = p;
    setLink(pa->p, this, pa->dir());
    setLink(pa, c[x^1], x); setLink(this, pa, x^1);
    pa->calc();
}
void splay() {
    while (!isRoot() && !p->isRoot()) {
        p->p->prop(), p->prop(), prop();
        dir() == p->dir() ? p->rot() : rot();
        rot();
    }
    if (!isRoot()) p->prop(), prop(), rot();
    prop(); calc();
}
sn fbo(int b) { // find by order
    prop(); int z = getSz(c[0]); // of splay tree
    if (b == z) { splay(); return this; }
    return b < z ? c[0]->fbo(b) : c[1] -> fbo(b-z-1);
}
////////// BASE OPERATIONS
void access() { // bring this to top of tree, propagate
    for (sn v = this, pre = NULL; v; v = v->p) {
        v->splay(); // now switch virtual children
        if (pre) v->vsub -= pre->sub;
        if (v->c[1]) v->vsub += v->c[1]->sub;
        v->c[1] = pre; v->calc(); pre = v;
    }
    splay(); assert(!c[1]); // right subtree is empty
}
void makeRoot() {
    access(); flip ^= 1; access(); assert(!c[0] && !c[1]); }
////////// QUERIES
friend sn lca(sn x, sn y) {
    if (x == y) return x;
    x->access(), y->access(); if (!x->p) return NULL;
    x->splay(); return x->p?:x; // y was below x in latter case
} // access at y did not affect x -> not connected
friend bool connected(sn x, sn y) { return lca(x,y); }
// # nodes above
int distRoot() { access(); return getSz(c[0]); }
sn getRoot() { // get root of LCT component
    access(); sn a = this;
    while (a->c[0]) a = a->c[0], a->prop();
    a->access(); return a;
}
sn getPar(int b) { // get b-th parent on path to root
    access(); b = getSz(c[0])-b; assert(b >= 0);
    return fbo(b);
```

```
} // can also get min, max on path to root, etc
////////// MODIFICATIONS
void set(int v) { access(); val = v; calc(); }
friend void link(sn x, sn y, bool force = 0) {
    assert(!connected(x,y));
    if (force) y->makeRoot(); // make x par of y
    else { y->access(); assert(!y->c[0]); }
    x->access(); setLink(y,x,0); y->calc();
}
friend void cut(sn y) { // cut y from its parent
    y->access(); assert(y->c[0]);
    y->c[0]->p = NULL; y->c[0] = NULL; y->calc(); }
friend void cut(sn x, sn y) { // if x, y adj in tree
    x->makeRoot(); y->access();
    assert(y->c[0] == x && !x->c[0] && !x->c[1]); cut(y); }
};
sn LCT[MX];
```

```
////////// THE APPLICANT SOLUTION
void setNex(sn a, sn b) { // set f[a] = b
    if (connected(a,b)) a->extra = b;
    else link(b,a); }
void delNex(sn a) { // set f[a] = NULL
    auto t = a->getRoot();
    if (t == a) { t->extra = NULL; return; }
    cut(a); assert(t->extra);
    if (!connected(t,t->extra))
        link(t->extra,t), t->extra = NULL;
}
sn getPar(sn a, int b) { // get f^b[a]
    int d = a->distRoot(); if (b <= d) return a->getPar(b);
    b -= d+1; auto r = a->getRoot()->extra; assert(r);
    d = r->distRoot()+1; return r->getPar(b%d);
}
```

## Geometry (8)

### 8.1 Primitives

|                                                                                                                                                                   |                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| ComplexComp.h                                                                                                                                                     |                 |
| <b>Description:</b> Allows you to sort complex numbers.                                                                                                           | 6f828b, 5 lines |
| <pre>#define x real() #define y imag() using P = complex&lt;db&gt;; namespace std {     bool operator&lt;(P l,P r) { return mp(l.x,l.y)&lt;mp(r.x,r.y); } }</pre> |                 |

|                                                 |                  |
|-------------------------------------------------|------------------|
| PointShort.h                                    |                  |
| <b>Description:</b> Use in place of complex<T>. | 268ecf, 36 lines |

```
using T = db; // or ll
const T EPS = 1e-9; // adjust as needed
using P = pair<T,T>; using vP = V<P>; using Line = pair<P,P>;
int sgn(T a) { return (a>EPS)-(a<-EPS); }
T sq(T a) { return a*a; }

T abs2(P p) { return sq(p.f)+sq(p.s); }
T abs(P p) { return sqrt(abs2(p)); }
T arg(P p) { return atan2(p.s,p.f); }
P conj(P p) { return P(p.f,-p.s); }
P perp(P p) { return P(-p.s,p.f); }
P dir(T ang) { return P(cos(ang),sin(ang)); }

P operator+(P l, P r) { return P(l.f+r.f,l.s+r.s); }
P operator-(P l, P r) { return P(l.f-r.f,l.s-r.s); }
P operator*(P l, T r) { return P(l.f*r,l.s*r); }
P operator/(P l, T r) { return P(l.f/r,l.s/r); }
```

```
P operator*(P l, P r) { // complex # multiplication
    return P(l.f*r.f-l.s*r.s,l.s*r.f+l.f*r.s); }
P operator/(P l, P r) { return l*conj(r)/abs2(r); }

P unit(const P& p) { return p/abs(p); }
T dot(const P& a, const P& b) { return a.f*b.f+a.s*b.s; }
T dot(const P& p, const P& a, const P& b) { return dot(a-p,b-p)
    ↪; }
T cross(const P& a, const P& b) { return a.f*b.s-a.s*b.f; }
T cross(const P& p, const P& a, const P& b) {
    return cross(a-p,b-p); }
P reflect(const P& p, const Line& l) {
    P a = l.f, d = l.s-l.f;
    return a+conj((p-a)/d)*d; }
P foot(const P& p, const Line& l) {
    return (p+reflect(p,l))/(T)2; }
bool onSeg(const P& p, const Line& l) {
    return sgn(cross(l.f,l.s,p)) == 0 && sgn(dot(p,l.f,l.s)) <= 0
    ↪; }
ostream& operator<<(ostream& os, P p) {
    return os << "(" << p.f << ", " << p.s << ")"; }
```

AngleCmp.h  
**Description:** Sorts points in ccw order about origin in the same way as atan2, which returns real in  $(-\pi,\pi]$  so points on negative  $x$ -axis come last.  
**Usage:** vP v; sort(all(v),angleCmp);

|                                                                                                                                                  |                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| "Point.h"                                                                                                                                        | 2df5fc, 8 lines |
| // WARNING: you will get unexpected results if you mistype this<br>↪ as bool instead of int<br>// -1 if lower half, 0 if origin, 1 if upper half |                 |
| int half(P x) { return x.s != 0 ? sgn(x.s) : -sgn(x.f); }                                                                                        |                 |
| bool angleCmp(P a, P b) { int A = half(a), B = half(b);<br>return A == B ? cross(a,b) > 0 : A < B; }                                             |                 |
| <br>// equivalent to: sort(all(v),[](P a, P b) {<br>// return atan2(a.s,a.f) < atan2(b.s,b.f); });                                               |                 |

SegDist.h  
**Description:** computes distance between  $P$  and line (segment)  $AB$

|                                                                                                                                                                 |                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| "Point.h"                                                                                                                                                       | 0cb69a, 6 lines |
| T lineDist(const P& p, const Line& l) {<br>return abs(cross(p,l.f,l.s))/abs(l.f-l.s); }                                                                         |                 |
| T segDist(const P& p, const Line& l) {<br>if (dot(l.f,p,l.s) <= 0) return abs(p-l.f);<br>if (dot(l.s,p,l.f) <= 0) return abs(p-l.s);<br>return lineDist(p,l); } |                 |

SegIsect.h  
**Description:** computes the intersection point(s) of line (segments)  $a$  and  $b$

|                                                                                                                                                                                                                                                                    |                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| "Point.h"                                                                                                                                                                                                                                                          | 7f6ba0, 26 lines |
| // {unique intersection point} if it exists<br>// {b.f,b.s} if input lines are the same<br>// empty if lines do not intersect                                                                                                                                      |                  |
| vP lineIsect(const Line& a, const Line& b) {<br>T a0 = cross(a.f,a.s,b.f), a1 = cross(a.f,a.s,b.s);<br>if (a0 == a1) return a0 == 0 ? vP{b.f,b.s} : vP{};<br>return {(b.s*a0-b.f*a1)/(a0-a1)};                                                                     |                  |
| }                                                                                                                                                                                                                                                                  |                  |
| <br>// point in interior of both segments a and b, if it exists                                                                                                                                                                                                    |                  |
| vP strictIsect(const Line& a, const Line& b) {<br>T a0 = cross(a.f,a.s,b.f), a1 = cross(a.f,a.s,b.s);<br>T b0 = cross(b.f,b.s,a.f), b1 = cross(b.f,b.s,a.s);<br>if (sgn(a0)*sgn(a1) < 0 && sgn(b0)*sgn(b1) < 0)<br>return {(b.s*a0-b.f*a1)/(a0-a1)};<br>return {}; |                  |
| }                                                                                                                                                                                                                                                                  |                  |

```
// intersection of segments, a and b may be degenerate
vP segIsect(const Line& a, const Line& b) {
    vP v = strictIsect(a,b); if (sz(v)) return v;
    set<P> s;
    #define i(x,y) if (onSeg(x,y)) s.ins(x)
    i(a.f,b); i(a.s,b); i(b.f,a); i(b.s,a);
    return {all(s)};
}
```

## 8.2 Polygons

PolygonCenArea.h  
**Description:** centroid (center of mass) of a polygon with constant mass per unit area and SIGNED area  
**Time:**  $\mathcal{O}(N)$

|           |                 |
|-----------|-----------------|
| "Point.h" | 4ca221, 7 lines |
|-----------|-----------------|

```
pair<P,T> cenArea(const vP& v) { assert(sz(v) >= 3);
    P cen{}; T area{};
    F0R(i,sz(v)) {
        int j = (i+1)%sz(v); T a = cross(v[i],v[j]);
        cen += a*(v[i]+v[j]); area += a; }
    return {cen/area/(T)3,area/2}; // area is SIGNED
}
```

InPolygon.h  
**Description:** Tests whether point is inside, on, or outside of a polygon (returns -1, 0, or 1). Both CW and CCW polygons are ok.  
**Time:**  $\mathcal{O}(N)$

|                                                                                                                                                                                                                                                                                 |                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| "Point.h"                                                                                                                                                                                                                                                                       | 6ebff0, 9 lines |
| int inPoly(const P& p, const vP& poly) {<br>int n = sz(poly), ans = 0;<br>F0R(i,n) {<br>P x = poly[i], y = poly[(i+1)%n]; if (x.s > y.s) swap(x,y);<br>if (onSeg(p,{x,y})) return 0;<br>ans ^= (x.s <= p.s && p.s < y.s && cross(p,x,y) > 0);<br>}<br>return ans ? -1 : 1;<br>} |                 |

ConvexHull.h  
**Description:** top-bottom convex hull  
**Time:**  $\mathcal{O}(N \log N)$

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| "Point.h"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 868655, 18 lines |
| pair<vi,vi> ulHull(const vP& v) {<br>vi p(sz(v)), u, l; iota(all(p), 0);<br>sort(all(p), [&v](int a, int b) { return v[a] < v[b]; });<br>each(i,p) {<br>#define ADDP(C, cmp) while (sz(C) > 1 && cross(\<br>v[C[sz(C)-2]],v[C.bk],v[i]) cmp 0) C.pop_back(); C.pb(i);<br>ADDP(u, >=); ADDP(l, <=);<br>}<br>return {u,l};<br>}<br>vi hullInd(const vP& v) { // returns indices in CCW order<br>vi u,l; tie(u,l) = ulHull(v); if (sz(l) <= 1) return l;<br>if (v[l[0]] == v[l[1]]) return {0};<br>l.insert(end(l),l+rall(u)-1); return l;<br>}<br>vP hull(const vP& v) {<br>vi w = hullInd(v); vP res; each(t,w) res.pb(v[t]);<br>return res; } |                  |

MinkowskiSum.h  
**Description:** Minkowski sum of two convex polygons given in CCW order.  
**Time:**  $\mathcal{O}(N)$

|                |                  |
|----------------|------------------|
| "ConvexHull.h" | e5d935, 29 lines |
|----------------|------------------|

```
vP minkowski_sum(vP a, vP b) {
    if (sz(a) > sz(b)) swap(a, b);
    if (!sz(a)) return {};
    if (sz(a) == 1) {
        each(t, b) t += a.ft;
        return b;
    }
    rotate(begin(a), min_element(all(a)), end(a));
    rotate(begin(b), min_element(all(b)), end(b));
    a.pb(a[0]), a.pb(a[1]);
    b.pb(b[0]), b.pb(b[1]);
    vP result;
    int i = 0, j = 0;
    while (i < sz(a)-2 || j < sz(b)-2) {
        result.pb(a[i]+b[j]);
        T crs = cross(a[i+1]-a[i],b[j+1]-b[j]);
        i += (crs >= 0);
        j += (crs <= 0);
    }
    return result;
}
```

T diameter2(vP p) { // example application: squared diameter  
vP a = hull(p);  
vP b = a; each(t, b) t \*= -1;  
vP c = minkowski\_sum(a, b);  
T ret = 0; each(t, c) ckmx(ret, abs2(t));  
return ret;  
}

HullDiameter.h  
**Description:** Rotating calipers. Returns square of greatest distance between two points in  $P$ .  
**Time:**  $\mathcal{O}(N)$  given convex hull

|                                                                                                                                                                                                                                                               |                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| "ConvexHull.h"                                                                                                                                                                                                                                                | 213a2a, 9 lines |
| db diameter2(vP P) {<br>P = hull(P);<br>int n = sz(P), ind = 1; T ans = 0;<br>if (n > 1) F0R(i,n) for (int j = (i+1)%n;;ind = (ind+1)%n) {<br>ckmx(ans, abs2(P[i]-P[ind]));<br>if (cross(P[j]-P[i],P[(ind+1)%n]-P[ind]) <= 0) break;<br>}<br>return ans;<br>} |                 |

HullTangents.h  
**Description:** Given convex polygon with no three points collinear and a point strictly outside of it, computes the lower and upper tangents.  
**Time:**  $\mathcal{O}(\log N)$

|                                                                                                                                                                                                                                                                                                                                                                                                                                |                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| "Point.h"                                                                                                                                                                                                                                                                                                                                                                                                                      | 85b807, 36 lines |
| bool lower;<br>bool better(P a, P b, P c) {<br>T z = cross(a,b,c);<br>return lower ? z < 0 : z > 0; }<br>int tangent(const vP& a, P b) {<br>if (sz(a) == 1) return 0;<br>int lo, hi;<br>if (better(b,a[0],a[1])) {<br>lo = 0, hi = sz(a)-1;<br>while (lo < hi) {<br>int mid = (lo+hi+1)/2;<br>if (better(b,a[0],a[mid])) lo = mid;<br>else hi = mid-1;<br>}<br>lo = 0;<br>} else {<br>lo = 1, hi = sz(a);<br>while (lo < hi) { |                  |



```
int mid = (lo+hi)/2;
if (!better(b,a[0],a[mid])) lo = mid+1;
else hi = mid;
}
hi = sz(a);
}
while (lo < hi) {
int mid = (lo+hi)/2;
if (better(b,a[mid],a[(mid+1)%sz(a)])) lo = mid+1;
else hi = mid;
}
return lo%sz(a);
}
pi tangents(const vP& a, P b) {
lower = 1; int x = tangent(a,b);
lower = 0; int y = tangent(a,b);
return {x,y};
}
```

LineHull.h  
Description: lineHull accepts line and ccw convex polygon. If all vertices in poly lie to one side of the line, returns a vector of closest vertices to line as well as orientation of poly with respect to line ( $\pm 1$  for above/below). Otherwise, returns the range of vertices that lie on or below the line. extrVertex returns the point of a hull with the max projection onto a line.  
Time:  $\mathcal{O}(\log N)$

```
"Point.h" 40e5a6, 41 lines
using Line = AR<P,2>;
#define cmp(i,j) sgn(-dot(dir,poly[(i)%n]-poly[(j)%n]))
#define extr(i) cmp(i+1,i) >= 0 && cmp(i,i-1+n) < 0
int extrVertex(const vP& poly, P dir) {
int n = sz(poly), lo = 0, hi = n;
if (extr(0)) return 0;
while (lo+1 < hi) {
int m = (lo+hi)/2;
if (extr(m)) return m;
int ls = cmp(lo+1,lo), ms = cmp(m+1,m);
(ls < ms || (ls == ms && ls == cmp(lo,m)) ? hi : lo) = m;
}
return lo;
}
vi same(Line line, const vP& poly, int a) {
// points on same parallel as a
int n = sz(poly); P dir = perp(line[0]-line[1]);
if (cmp(a+n-1,a) == 0) return {(a+n-1)%n,a};
if (cmp(a,a+1) == 0) return {a,(a+1)%n};
return {a};
}
#define cmpL(i) sgn(cross(line[0],line[1],poly[i]))
pair<int,vi> lineHull(Line line, const vP& poly) {
int n = sz(poly); assert(n>1);
int endA = extrVertex(poly,perp(line[0]-line[1])); // lowest
if (cmpL(endA) >= 0) return {1,same(line,poly,endA)};
int endB = extrVertex(poly,perp(line[1]-line[0])); // highest
if (cmpL(endB) <= 0) return {-1,same(line,poly,endB)};
AR<int,2> res;
F0R(i,2) {
int lo = endA, hi = endB; if (hi < lo) hi += n;
while (lo < hi) {
int m = (lo+hi+1)/2;
if (cmpL(m%n) == cmpL(endA)) lo = m;
else hi = m-1;
}
res[i] = lo%n; swap(endA,endB);
}
if (cmpL((res[0]+1)%n) == 0) res[0] = (res[0]+1)%n;
return {0,{(res[1]+1)%n,res[0]}};
}
```

HalfPlaneIsect.h  
Description: Returns vertices of half-plane intersection. A half-plane is the area to the left of a ray, which is defined by a point p and a direction dp. Area of intersection should be sufficiently precise when all inputs are integers with magnitude  $\leq 10^5$ . Intersection must be bounded. Probably works with floating point too (but EPS might need to be adjusted?).  
Time:  $\mathcal{O}(N \log N)$

```
"AngleCmp.h" 18f712, 52 lines
struct Ray {
P p, dp; // origin, direction
P isect(const Ray& L) const {
return p+dp*(cross(L.dp,L.p-p)/cross(L.dp,dp)); }
bool operator<(const Ray& L) const {
return angleCmp(dp,L.dp); }
};

vP halfPlaneIsect(V<Ray> rays, bool add_bounds = false) {
if (add_bounds) { // bound input by rectangle [0,DX] x [0,DY]
int DX = 1e9, DY = 1e9;
rays.pb({P{0,0},P{1,0}});
rays.pb({P{DX,0},P{0,1}});
rays.pb({P{DX,DY},P{-1,0}});
rays.pb({P{0,DY},P{0,-1}});
}
sor(rays); // sort rays by angle
{ // remove parallel rays
V<Ray> nrays;
each(t,rays) {
if (!sz(nrays) || cross(nrays.bk.dp,t.dp) > EPS) { nrays.
↳pb(t); continue; }
// last two rays are parallel, keep only one
if (cross(t.dp,t.p-nrays.bk.p) > 0) nrays.bk = t;
}
swap(rays, nrays);
}
auto bad = [&](const Ray& a, const Ray& b, const Ray& c) {
P p1 = a.isect(b), p2 = b.isect(c);
if (dot(p2-p1,b.dp) <= EPS) {
if (cross(a.dp,c.dp) <= 0) return 2; // isect(a,b,c) =
↳empty
return 1; // isect(a,c) == isect(a,b,c)
}
return 0; // all three rays matter
};
#define reduce(t) \
while (sz(poly) > 1) { \
int b = bad(poly.at(sz(poly)-2),poly.bk,t); \
if (b == 2) return {}; \
if (b == 1) poly.pop_back(); \
else break; \
}
deque<Ray> poly;
each(t,rays) { reduce(t); poly.pb(t); }
for(;;poly.pop_front()) {
reduce(poly[0]);
if (!bad(poly.bk,poly[0],poly[1])) break;
}
assert(sz(poly) >= 3); // expect nonzero area
vP poly_points; F0R(i,sz(poly))
poly_points.pb(poly[i].isect(poly[(i+1)%sz(poly)]));
return poly_points;
}
```

HalfPlaneSet.h  
Description: Online Half-Plane Intersection

```
d2027d, 77 lines
using T = int;
using T2 = long long;
using T4 = __int128_t;
```

```
const T2 INF = 2e9;

struct Line {
T a, b;
T2 c;
};

bool operator<(Line m, Line n) {
auto half = [&](Line m) { return m.b < 0 || m.b == 0 && m.a <
↳0; };
return make_tuple(half(m), (T2)m.b * n.a) <
make_tuple(half(n), (T2)m.a * n.b);
}

tuple<T4, T4, T2> LineIntersection(Line m, Line n) {
T2 d = (T2)m.a * n.b - (T2)m.b * n.a; // assert(d);
T4 x = (T4)m.c * n.b - (T4)m.b * n.c;
T4 y = (T4)m.a * n.c - (T4)m.c * n.a;
return {x, y, d};
}

Line LineFromPoints(T x1, T y1, T x2, T y2) {
// everything to the right of ray {x1, y1} -> {x2, y2}
T a = y1 - y2, b = x2 - x1;
T2 c = (T2)a * x1 + (T2)b * y1;
return {a, b, c}; // ax + by <= c
}

ostream &operator<<(ostream &out, Line l) {
out << "Line " << l.a << " " << l.b << " " << -l.c;
// out << "(" << l.a << " * x + " << l.b << " * y <= " << l.c
↳<< ")";
return out;
}

struct HalfplaneSet : multiset<Line> {
HalfplaneSet() {
insert({+1, 0, INF});
insert({0, +1, INF});
insert({-1, 0, INF});
insert({0, -1, INF});
};
auto adv(auto it, int z) { // z = {-1, +1}
return (z == -1 ? --(it == begin() ? end() : it)
: (++it == end() ? begin() : it));
}

bool chk(auto it) {
Line l = *it, pl = *adv(it, -1), nl = *adv(it, +1);
auto [x, y, d] = LineIntersection(pl, nl);
T4 sat = l.a * x + l.b * y - (T4)l.c * d;
if (d < 0 && sat < 0) return clear(), 0; // unsat
if ((d > 0 && sat <= 0) || (d == 0 && sat < 0)) return
↳erase(it), 1;
return 0;
}

void Cut(Line l) { // add ax + by <= c
if (empty()) return;
auto it = insert(l);
if (chk(it)) return;
for (int z : {-1, +1})
while (size() && chk(adv(it, z)))
;
}

double Maximize(T a, T b) { // max ax + by (UNTESTED)
if (empty()) return -1 / 0.;
auto it = lower_bound({a, b});
if (it == end()) it = begin();
auto [x, y, d] = LineIntersection(*adv(it, -1), *it);
return (l.0 * a * x + l.0 * b * y) / d;
}

double Area() {
double total = 0.;
}
```

```
for (auto it = begin(); it != end(); ++it) {
    auto [x1, y1, d1] = LineIntersection(*adv(it, -1), *it);
    auto [x2, y2, d2] = LineIntersection(*it, *adv(it, +1));
    total += (1.0 * x1 * y2 - 1.0 * x2 * y1) / d1 / d2;
}
return total * 0.5;
}
```

PolygonUnion.kt

**Description:** Compute union or intersection of two polygons and compute the area of the resulting figure.

**Time:** Runs quite quickly for two convex polygons with  $10^5$  vertices each.

```
import java.awt.geom.* // at beginning of file
fun loadPoly(): DoubleArray {
    val n = rInt() // read n points
    return rDbs(2*n).toDoubleArray()
}
fun makeArea(ps: DoubleArray): Area {
    val p = Path2D.Double()
    p.moveTo(ps[0],ps[1])
    for (i in 0..ps.size/2-1) p.lineTo(ps[2*i],ps[2*i+1])
    p.closePath()
    return Area(p)
}
fun computeArea(a: Area): Double {
    val iter = a.getPathIterator(null)
    val buf = DoubleArray(6,{0.0})
    var ret = 0.0
    val ps = ArrayList<Double>()
    while (!iter.isDone()) {
        when(iter.currentSegment(buf)) {
            PathIterator.SEG_MOVETO,PathIterator.SEG_LINETO->{
                ps.add(buf[0])
                ps.add(buf[1])
            }
            PathIterator.SEG_CLOSE -> {
                ps.add(ps[0])
                ps.add(ps[1])
                for (i in 0..ps.size/2-2) ret -= ps[2*i]*ps[2*i+1]
                ps.clear()
            }
        }
        iter.next()
    }
    assert(ret >= 0)
    return ret/2
}
fun intersectArea(a: DoubleArray, b: DoubleArray): Double {
    val ret = makeArea(a)
    ret.intersect(makeArea(b)) // or .add for union
    return computeArea(ret)
}
```

8.3 Circles

Circle.h

**Description:** represent circle as {center,radius}

```
"Point.h" 91f3fc, 6 lines
using Circ = pair<P,T>;
int in(const Circ& x, const P& y) { // -1 if inside, 0, 1
    return sgn(abs(y-x.f)-x.s); }
T arclength(const Circ& x, P a, P b) {
    // precondition: a and b on x
    P d = (a-x.f)/(b-x.f); return x.s*acos(d.f); }
```

CircleIsect.h

**Description:** Circle intersection points and intersection area. Tangents will be returned twice.

```
"Circle.h" 21a173, 22 lines
vP isect(const Circ& x, const Circ& y) { // precondition: x!=y
    T d = abs(x.f-y.f), a = x.s, b = y.s;
    if (sgn(d) == 0) { assert(a != b); return {}; }
    T C = (a*a+d*d-b*b)/(2*a*d);
    if (abs(C) > 1+EPS) return {};
    T S = sqrt(max(1-C*C,(T)0)); P tmp = (y.f-x.f)/d*x.s;
    return {x.f+tmp*P(C,S),x.f+tmp*P(C,-S)};
}
vP isect(const Circ& x, const Line& y) {
    P c = foot(x.f,y); T sq_dist = sq(x.s)-abs2(x.f-c);
    if (sgn(sq_dist) < 0) return {};
    P offset = unit(y.s-y.f)*sqrt(max(sq_dist,T(0)));
    return {c+offset,c-offset};
}
T isect_area(Circ x, Circ y) { // not thoroughly tested
    T d = abs(x.f-y.f), a = x.s, b = y.s; if (a < b) swap(a,b);
    if (d >= a+b) return 0;
    if (d <= a-b) return PI*b*b;
    T ca = (a*a+d*d-b*b)/(2*a*d), cb = (b*b+d*d-a*a)/(2*b*d);
    T s = (a+b+d)/2, h = 2*sqrt(s*(s-a)*(s-b)*(s-d))/d;
    return a*a*acos(ca)+b*b*acos(cb)-d*h;
}
```

CircleTangents.h

**Description:** internal and external tangents between two circles

```
"Circle.h" d9a76f, 22 lines
P tangent(P x, Circ y, int t = 0) {
    y.s = abs(y.s); // abs needed because internal calls y.s < 0
    if (y.s == 0) return y.f;
    T d = abs(x-y.f);
    P a = pow(y.s/d,2)*(x-y.f)+y.f;
    P b = sqrt(d*d-y.s*y.s)/d*y.s*unit(x-y.f)*dir(PI/2);
    return t == 0 ? a+b : a-b;
}
V<pair<P,P>> external(Circ x, Circ y) {
    V<pair<P,P>> v;
    if (x.s == y.s) {
        P tmp = unit(x.f-y.f)*x.s*dir(PI/2);
        v.eb(x.f+tmp,y.f+tmp);
        v.eb(x.f-tmp,y.f-tmp);
    } else {
        P p = (y.s*x.f-x.s*y.f)/(y.s-x.s);
        F0R(i,2) v.eb(tangent(p,x,i),tangent(p,y,i));
    }
    return v;
}
V<pair<P,P>> internal(Circ x, Circ y) {
    return external({x.f,-x.s},y); }
```

Circumcenter.h

**Description:** returns {circumcenter,circumradius}

```
"Circle.h" a2c6a6, 5 lines
Circ ccCenter(P a, P b, P c) {
    b -= a; c -= a;
    P res = b*c*(conj(c)-conj(b))/(b*conj(c)-conj(b)*c);
    return {a+res,abs(res)};
}
MinEnclosingCirc.h
Description: minimum enclosing circle
Time: expected  $\mathcal{O}(N)$ 
"Circumcenter.h" 53963d, 13 lines
circ mec(vP ps) {
```

```
shuffle(all(ps), rng);
P o = ps[0]; T r = 0, EPS = 1+1e-8;
F0R(i,sz(ps)) if (abs(o-ps[i]) > r*EPS) {
    o = ps[i], r = 0; // point is on MEC
    F0R(j,i) if (abs(o-ps[j]) > r*EPS) {
        o = (ps[i]+ps[j])/2, r = abs(o-ps[i]);
        F0R(k,j) if (abs(o-ps[k]) > r*EPS)
            tie(o,r) = ccCenter(ps[i],ps[j],ps[k]);
    }
}
return {o,r};
}
```

8.4 Misc

ClosestPair.h

**Description:** Line sweep to find two closest points .

**Time:**  $\mathcal{O}(N \log N)$

```
"Point.h" 2b60fa, 17 lines
pair<P,P> solve(vP v) {
    pair<db,pair<P,P>> bes; bes.f = INF;
    set<P> S; int ind = 0;
    sort(all(v));
    F0R(i,sz(v)) {
        if (i && v[i] == v[i-1]) return {v[i],v[i]};
        for (; v[i].f-v[ind].f >= bes.f; ++ind)
            S.erase({v[ind].s,v[ind].f});
        for (auto it = S.sub({v[i].s-bes.f,INF});
            it != end(S) && it->f < v[i].s+bes.f; ++it) {
            P t = {it->s,it->f};
            ckmin(bes,{abs(t-v[i]),{t,v[i]}});
        }
        S.insert({v[i].s,v[i].f});
    }
    return bes.s;
}
```

DelaunayFast.h

**Description:** Fast Delaunay triangulation assuming no duplicates and not all points collinear (in latter case, result will be empty). Should work for doubles as well, though there may be precision issues in 'circ'. Returns triangles in ccw order. Each circumcircle will contain none of the input points. If coordinates are ints at most  $B$  then T should be large enough to support ints on the order of  $B^4$ .

**Time:**  $\mathcal{O}(N \log N)$

```
"Point.h" 7ff542, 82 lines
// using l1l = ll; (if coords are < 2e4)
using l1l = __int128;
// returns true if p strictly within circumcircle(a,b,c)
bool inCircle(P p, P a, P b, P c) {
    a -= p, b -= p, c -= p; // assert(cross(a,b,c)>0);
    l1l x = (l1l)abs2(a)*cross(b,c)+(l1l)abs2(b)*cross(c,a)
        +(l1l)abs2(c)*cross(a,b);
    return x*(cross(a,b,c)>0?1:-1) > 0;
}
P arb(LLONG_MAX,LLONG_MAX); // not equal to any other point
using Q = struct Quad*;
struct Quad {
    bool mark; Q o, rot; P p;
    P F() { return r()->p; }
    Q r() { return rot->rot; }
    Q prev() { return rot->o->rot; }
    Q next() { return r()->prev(); }
};
Q makeEdge(P orig, P dest) {
    Q q[]{new Quad{0,0,0,orig}, new Quad{0,0,0,arb},
        new Quad{0,0,0,dest}, new Quad{0,0,0,arb}};
    F0R(i,4) q[i]->o = q[-i & 3], q[i]->rot = q[(i+1) & 3];
```

```
    return *q;
}

void splice(Q a, Q b) { swap(a->o->rot->o, b->o->rot->o); swap(
    ↪a->o, b->o); }
Q connect(Q a, Q b) {
    Q q = makeEdge(a->F(), b->p);
    splice(q, a->next()); splice(q->r(), b);
    return q;
}
pair<Q,Q> rec(const vP& s) {
    if (sz(s) <= 3) {
        Q a = makeEdge(s[0], s[1]), b = makeEdge(s[1], s.bk);
        if (sz(s) == 2) return { a, a->r() };
        splice(a->r(), b);
        auto side = cross(s[0], s[1], s[2]);
        Q c = side ? connect(b, a) : 0;
        return {side < 0 ? c->r() : a, side < 0 ? c : b->r() };
    }

#define H(e) e->F(), e->p
#define valid(e) (cross(e->F(),H(base)) > 0)
    Q A, B, ra, rb;
    int half = sz(s) / 2;
    tie(ra, A) = rec({all(s)-half});
    tie(B, rb) = rec({sz(s)-half+all(s)});
    while ((cross(B->p,H(A)) < 0 && (A = A->next())) ||
        (cross(A->p,H(B)) > 0 && (B = B->r()->o)));
    Q base = connect(B->r(), A);
    if (A->p == ra->p) ra = base->r();
    if (B->p == rb->p) rb = base;

#define DEL(e, init, dir) Q e = init->dir; if (valid(e)) \
    while (inCircle(e->dir->F(), H(base), e->F())) { \
        Q t = e->dir; \
        splice(e, e->prev()); \
        splice(e->r(), e->r()->prev()); \
        e = t; \
    }
    while (1) {
        DEL(LC, base->r(), o); DEL(RC, base, prev());
        if (!valid(LC) && !valid(RC)) break;
        if (!valid(LC) || (valid(RC) && inCircle(H(RC), H(LC))))
            base = connect(RC, base->r());
        else base = connect(base->r(), LC->r());
    }
    return {ra, rb};
}
V<AR<P,3>> triangulate(vP pts) {
    sor(pts); assert(unique(all(pts)) == end(pts)); // no
    ↪duplicates
    if (sz(pts) < 2) return {};
    Q e = rec(pts).f; V<Q> q = {e};
    while (cross(e->o->F(), e->F(), e->p) < 0) e = e->o;
#define ADD { Q c = e; do { c->mark = 1; pts.pb(c->p); \
    q.pb(c->r()); c = c->next(); } while (c != e); }
    ADD; pts.clear();
    int qi = 0; while (qi < sz(q)) if (!(e = q[qi++])->mark) ADD;
    V<AR<P,3>> ret(sz(pts)/3);
    F0R(i,sz(pts)) ret[i/3][i%3] = pts[i];
    return ret;
}
}
```

ManhattanMST.h

**Description:** Given  $N$  points, returns up to  $4N$  edges which are guaranteed to contain a MST for graph with edge weights  $w(p,q) = |p.x-q.x|+|p.y-q.y|$ . Edges are in the form {dist, {src, dst}}.

**Time:**  $\mathcal{O}(N \log N)$

```
"DSU.h" b7a3bd, 24 lines
// use standard MST algorithm on result to find final MST
```

```
V<pair<int,pi>> manhattanMst(vpi v) {
    vi id(sz(v)); iota(all(id),0);
    V<pair<int,pi>> ed;
    F0R(k,4) {
        sort(all(id), [&](int i, int j) {
            return v[i].f+v[i].s < v[j].f+v[j].s; });
        map<int,int> sweep; // find first octant neighbors
        each(i,id) { // those in sweep haven't found neighbor yet
            for (auto it = sweep.lb(-v[i].s);
                it != end(sweep); sweep.erase(it++)) {
                int j = it->s;
                pi d{v[i].f-v[j].f,v[i].s-v[j].s};if (d.s>d.f)break;
                ed.pb({d.f+d.s,{i,j}});
            }
            sweep[-v[i].s] = i;
        }
        each(p,v) {
            if (k&1) p.f *= -1;
            else swap(p.f,p.s);
        }
    }
    return ed;
}
```

8.5 3D

Point3D.h

**Description:** Basic 3D geometry.

```
"Point.h" d7d537, 82 lines
using P3 = AR<T,3>; using Tri = AR<P3,3>; using vP3 = V<P3>;
T abs2(const P3& x) {
    T sum = 0; F0R(i,3) sum += sq(x[i]);
    return sum; }
T abs(const P3& x) { return sqrt(abs2(x)); }

P3& operator+=(P3& l, const P3& r) { F0R(i,3) l[i] += r[i];
    return l; }
P3& operator-=(P3& l, const P3& r) { F0R(i,3) l[i] -= r[i];
    return l; }
P3& operator*=(P3& l, const T& r) { F0R(i,3) l[i] *= r;
    return l; }
P3& operator/=(P3& l, const T& r) { F0R(i,3) l[i] /= r;
    return l; }
P3 operator-(P3 l) { l *= -1; return l; }
P3 operator+(P3 l, const P3& r) { return l += r; }
P3 operator-(P3 l, const P3& r) { return l -= r; }
P3 operator*(P3 l, const T& r) { return l *= r; }
P3 operator*(const T& r, const P3& l) { return l*r; }
P3 operator/(P3 l, const T& r) { return l /= r; }
```

```
P3 unit(const P3& x) { return x/abs(x); }
T dot(const P3& a, const P3& b) {
    T sum = 0; F0R(i,3) sum += a[i]*b[i];
    return sum; }
P3 cross(const P3& a, const P3& b) {
    return {a[1]*b[2]-a[2]*b[1],a[2]*b[0]-a[0]*b[2],
        a[0]*b[1]-a[1]*b[0]}; }
P3 cross(const P3& a, const P3& b, const P3& c) {
    return cross(b-a,c-a); }
P3 perp(const P3& a, const P3& b, const P3& c) {
    return unit(cross(a,b,c)); }
```

```
bool isMult(const P3& a, const P3& b) { // for long longs
    P3 c = cross(a,b); F0R(i,sz(c)) if (c[i] != 0) return 0;
    return 1; }
bool collinear(const P3& a, const P3& b, const P3& c) {
    return isMult(b-a,c-a); }
```

```
T DC(const P3&a,const P3&b,const P3&c,const P3&p) {
```

```
    return dot(cross(a,b,c),p-a); }
bool coplanar(const P3&a,const P3&b,const P3&c,const P3&p) {
    return DC(a,b,c,p) == 0; }
bool op(const P3& a, const P3& b) {
    int ind = 0; // going in opposite directions?
    F0R(i,1,3) if (std::abs(a[i]*b[i])>std::abs(a[ind]*b[ind]))
        ind = i;
    return a[ind]*b[ind] < 0;
}
// coplanar points, b0 and b1 on opposite sides of a0-a1?
bool opSide(const P3&a,const P3&b,const P3&c,const P3&d) {
    return op(cross(a,b,c),cross(a,b,d)); }
// coplanar points, is a in Triangle b
bool inTri(const P3& a, const Tri& b) {
    F0R(i,3)if(opSide(b[i],b[(i+1)%3],b[(i+2)%3],a))return 0;
    return 1; }
```

```
// point-seg dist
T psDist(const P3&p,const P3&a,const P3&b) {
    if (dot(a-p,a-b) <= 0) return abs(a-p);
    if (dot(b-p,b-a) <= 0) return abs(b-p);
    return abs(cross(p,a,b))/abs(a-b);
}
// projection onto line
P3 foot(const P3& p, const P3& a, const P3& b) {
    P3 d = unit(b-a); return a+dot(p-a,d)*d; }
// rotate p about axis
P3 rotAxis(const P3& p, const P3& a, const P3& b, T theta) {
    P3 dz = unit(b-a), f = foot(p,a,b);
    P3 dx = p-f, dy = cross(dz,dx);
    return f+cos(theta)*dx+sin(theta)*dy;
}
// projection onto plane
P3 foot(const P3& a, const Tri& b) {
    P3 c = perp(b[0],b[1],b[2]);
    return a-c*(dot(a,c)-dot(b[0],c)); }
// line-plane intersection
P3 lpIntersect(const P3&a0,const P3&a1,const Tri&b) {
    P3 c = unit(cross(b[2]-b[0],b[1]-b[0]));
    T x = dot(a0,c)-dot(b[0],c), y = dot(a1,c)-dot(b[0],c);
    return (y*a0-x*a1)/(y-x);
}
```

Hull3D.h

**Description:** Incremental 3D convex hull where not all points are coplanar. Normals to returned faces point outwards. If coordinates are ints at most  $B$  then  $T$  should be large enough to support ints on the order of  $B^3$ . Changes order of points. The number of returned faces may depend on the random seed, because points that are on the boundary of the convex hull may or may not be included in the output.

**Time:**  $\mathcal{O}(N^2)$ ,  $\mathcal{O}(N \log N)$

```
// using T = ll;
bool above(const P3&a,const P3&b,const P3&c,const P3&p) {
    return DC(a,b,c,p) > 0; } // is p strictly above plane
void prep(vP3& p) { // rearrange points such that
    shuffle(all(p),rng); // first four are not coplanar
    int dim = 1;
    F0R(i,1,sz(p))
        if (dim == 1) {
            if (p[0] != p[i]) swap(p[1],p[i]), ++dim;
        } else if (dim == 2) {
            if (!collinear(p[0],p[1],p[i]))
                swap(p[2],p[i]), ++dim;
        } else if (dim == 3) {
            if (!coplanar(p[0],p[1],p[2],p[i]))
                swap(p[3],p[i]), ++dim;
        }
    assert(dim == 4);
```

```
}

using F = AR<int,3>; // face
V<F> hull3d(vP3& p) {
    // s.t. first four points form tetra
    prep(p); int N = sz(p); V<F> hull; // triangle for each face
    auto ad = [&](int a, int b, int c) { hull.pb({a,b,c}); };
    // +new face to hull
    ad(0,1,2), ad(0,2,1); // initialize hull as first 3 points
    V<vb> in(N,vb(N)); // is zero before each iteration
    FOR(i,3,N) { // incremental construction
        V<F> def, HULL; swap(hull,HULL);
        // HULL now contains old hull
        auto ins = [&](int a, int b, int c) {
            if (in[b][a]) in[b][a] = 0; // kill reverse face
            else in[a][b] = 1, ad(a,b,c);
        };
        each(f,HULL) {
            if (above(p[f[0]],p[f[1]],p[f[2]],p[i]))
                F0R(j,3) ins(f[j],f[(j+1)%3],i);
            // recalc all faces s.t. point is above face
            else def.pb(f);
        }
        each(t,hull) if (in[t[0]][t[1]]) // edge exposed,
            in[t[0]][t[1]] = 0, def.pb(t); // add a new face
        swap(hull,def);
    }
    return hull;
}
V<F> hull3dFast(vP3& p) {
    prep(p); int N = sz(p); V<F> hull;
    vb active; // whether face is active
    V<vi> rvis; // points visible from each face
    V<AR<pi,3>> other; // other face adjacent to each edge of
        ↪face
    V<vi> vis(N); // faces visible from each point
    auto ad = [&](int a, int b, int c) {
        hull.pb({a,b,c}); active.pb(1); rvis.eb(); other.eb(); };
    auto ae = [&](int a, int b) { vis[b].pb(a), rvis[a].pb(b); };
    auto abv = [&](int a, int b) {
        F f=hull[a]; return above(p[f[0]],p[f[1]],p[f[2]],p[b]); };
    auto edge = [&](pi e) -> pi {
        return {hull[e.f][e.s],hull[e.f][(e.s+1)%3]}; };
    auto glue = [&](pi a, pi b) { // link two faces by an edge
        pi x = edge(a); assert(edge(b) == mp(x.s,x.f));
        other[a.f][a.s] = b, other[b.f][b.s] = a;
    }; // ensure face 0 is removed when i=3
    ad(0,1,2), ad(0,2,1); if (abv(1,3)) swap(p[1],p[2]);
    F0R(i,3) glue({0,i},{1,2-i});
    FOR(i,3,N) ae(abv(1,i),i); // coplanar points go in rvis[0]
    vi label(N,-1);
    FOR(i,3,N) { // incremental construction
        vi rem; each(t,vis[i]) if (active[t]) active[t]=0, rem.pb(t
            ↪);
        if (!sz(rem)) continue; // hull unchanged
        int st = -1;
        each(r,rem) F0R(j,3) {
            int o = other[r][j].f;
            if (active[o]) { // create new face!
                int a,b; tie(a,b) = edge({r,j}); ad(a,b,i); st = a;
                int cur = sz(rvis)-1; label[a] = cur;
                vi tmp; set_union(all(rvis[r]),all(rvis[o]),
                    back_inserter(tmp));
                // merge sorted vectors ignoring duplicates
                each(x,tmp) if (abv(cur,x)) ae(cur,x);
                glue({cur,0},other[r][j]); // glue old w/ new face
            }
        }
        for (int x = st, y; ; x = y) { // glue new faces together
```

```
        int X = label[x]; glue({X,1},{label[y=hull[X][1]],2});
        if (y == st) break;
    }
}
V<F> ans; F0R(i,sz(hull)) if (active[i]) ans.pb(hull[i]);
return ans;
}
```

PolySaVol.h

Description: surface area and volume of polyhedron, normals to faces must point outwards

"Hull13D.h"52fc2b, 8 lines

pair<T,T> SaVol(vP3 p, V<F> faces) {  
 T s = 0, v = 0;  
 each(i,faces) {  
 P3 a = p[i[0]], b = p[i[1]], c = p[i[2]];  
 s += abs(cross(a,b,c)); v += dot(cross(a,b),c);  
 }  
 return {s/2,v/6};  
}

Strings (9)

9.1 Light

KMP.h

Description: f[i] is length of the longest proper suffix of the i-th prefix of s that is a prefix of s

Time: O(N)

4538e4, 13 lines

vi kmp(str s) {  
 int N = sz(s); vi f(N+1); f[0] = -1;  
 FOR(i,1,N+1) {  
 for (f[i]=f[i-1];f[i]!==-1&&s[f[i]]!=s[i-1];)f[i]=f[f[i]];  
 ++f[i]; }  
 return f;  
}  
vi getOc(str a, str b) { // find occurrences of a in b  
 vi f = kmp(a+"@"+b), ret;  
 FOR(i,sz(a),sz(b)+1) if (f[i+sz(a)+1] == sz(a))  
 ret.pb(i-sz(a));  
 return ret;  
}

Z.h

Description: f[i] is the max len such that s.substr(0,len) == s.substr(i,len)

Time: O(N)

566170, 15 lines

vi z(str s) {  
 int N = sz(s), L = 1, R = 0; s += '#';  
 vi ans(N); ans[0] = N;  
 FOR(i,1,N) {  
 if (i <= R) ans[i] = min(R-i+1,ans[i-L]);  
 while (s[i+ans[i]] == s[ans[i]]) ++ans[i];  
 if (i+ans[i]-1 > R) L = i, R = i+ans[i]-1;  
 }  
 return ans;  
}  
vi getPrefix(str a, str b) { // find prefixes of a in b  
 vi t = z(a+b); t = vi(sz(a)+all(t));  
 each(u,t) ckmn(u,sz(a));  
 return t;  
}

Manacher.h

Description: length of largest palindrome centered at each character of string and between every consecutive pair

Time: O(N)

fcc3f7, 13 lines

vi manacher(str \_S) {  
 str S = "@"; each(c,\_S) S += c, S += "#";  
 S.bk = '&';  
 vi ans(sz(S)-1); int lo = 0, hi = 0;  
 FOR(i,1,sz(S)-1) {  
 if (i != 1) ans[i] = min(hi-i,ans[hi-i+lo]);  
 while (S[i-ans[i]-1] == S[i+ans[i]+1]) ++ans[i];  
 if (i+ans[i] > hi) lo = i-ans[i], hi = i+ans[i];  
 }  
 ans.erase(begin(ans));  
 F0R(i,sz(ans)) if (i%2 == ans[i]%2) ++ans[i];  
 return ans;  
}

LyndonFactor.h

Description: A string is "simple" if it is strictly smaller than any of its own nontrivial suffixes. The Lyndon factorization of the string s is a factorization s = w1w2...wk where all strings wi are simple and w1 ≥ w2 ≥ ... ≥ wk.

Min rotation gets min index i such that cyclic shift of s starting at i is minimum.

Time: O(N)

af38ba, 19 lines

vs duval(str s) {  
 int N = sz(s); vs factors;  
 for (int i = 0; i < N; ) {  
 int j = i+1, k = i;  
 for (; j < N && s[k] <= s[j]; ++j) {  
 if (s[k] < s[j]) k = i;  
 else ++k;  
 }  
 for (; i <= k; i += j-k) factors.pb(s.substr(i,j-k));  
 }  
 return factors;  
}  
int minRotation(str s) {  
 int N = sz(s); s += s;  
 vs d = duval(s); int ind = 0, ans = 0;  
 while (ans+sz(d[ind]) < N) ans += sz(d[ind++]);  
 while (ind && d[ind] == d[ind-1]) ans -= sz(d[ind--]);  
 return ans;  
}

HashRange.h

Description: Polynomial hash for substrings with two bases.

fc0b90, 24 lines

using H = AR<int,2>; // bases not too close to ends  
H makeH(char c) { return {c,c}; }  
uniform\_int\_distribution<int> BDIST(0.1\*MOD,0.9\*MOD);  
const H base{BDIST(rng),BDIST(rng)};  
H operator+(H l, H r) {  
 F0R(i,2) if ((l[i] += r[i]) >= MOD) l[i] -= MOD;  
 return l; }  
H operator-(H l, H r) {  
 F0R(i,2) if ((l[i] -= r[i]) < 0) l[i] += MOD;  
 return l; }  
H operator\*(H l, H r) {  
 F0R(i,2) l[i] = (ll)l[i]\*r[i]%MOD;  
 return l; }  
  
V<H> pows{{1,1}};  
struct HashRange {  
 str S; V<H> cum{{}};  
 void add(char c) { S += c; cum.pb(base\*cum.bk+makeH(c)); }  
 void add(str s) { each(c,s) add(c); }  
 void extend(int len) { while (sz(pows) <= len)

```
    pows.pb(base*pows.bk); }
    H hash(int l, int r) { int len = r+1-l; extend(len);
      return cum[r+1]-pows[len]*cum[l]; }
};
```

ReverseBW.h

**Description:** Used only once. Burrows-Wheeler Transform appends # to a string, sorts the rotations of the string in increasing order, and constructs a new string that contains the last character of each rotation. This function reverses the transform.  
**Time:**  $\mathcal{O}(N \log N)$

```
str reverseBW(str t) {
    vi nex(sz(t)); iota(all(nex),0);
    stable_sort(all(nex), [&t](int a,int b){return t[a]<t[b];});
    str ret; for (int i = nex[0]; i; )
        ret += t[i = nex[i]];
    return ret;
}
```

e400d8, 7 lines

AhoCorasickFixed.h

**Description:** Aho-Corasick for fixed alphabet. For each prefix, stores link to max length suffix which is also a prefix.  
**Time:**  $\mathcal{O}(N \sum)$

96dfcc, 27 lines

```
template<size_t ASZ> struct ACfixed {
    struct Node { AR<int, ASZ> to; int link; };
    V<Node> d{{}};
    int add(str s) { // add word
        int v = 0;
        each(C,s) {
            int c = C-'a';
            if (!d[v].to[c]) d[v].to[c] = sz(d), d.eb();
            v = d[v].to[c];
        }
        return v;
    }
    void init() { // generate links
        d[0].link = -1;
        queue<int> q; q.push(0);
        while (sz(q)) {
            int v = q.ft; q.pop();
            F0R(c,ASZ) {
                int u = d[v].to[c]; if (!u) continue;
                d[u].link = d[v].link == -1 ? 0 : d[d[v].link].to[c];
                q.push(u);
            }
            if (v) F0R(c,ASZ) if (!d[v].to[c])
                d[v].to[c] = d[d[v].link].to[c];
        }
    }
};
```

SuffixArray.h

**Description:** Sort suffixes. First element of sa is sz(S), isa is the inverse of sa, and lcp stores the longest common prefix between every two consecutive elements of sa.  
**Time:**  $\mathcal{O}(N \log N)$

"RMQ.h" 27a566, 30 lines

```
struct SuffixArray {
    str S; int N; vi sa, isa, lcp;
    void init(str _S) { N = sz(S)+1; genSa(); genLcp(); }
    void genSa() { // sa has size sz(S)+1, starts with sz(S)
        sa = isa = vi(N); sa[0] = N-1; iota(1+all(sa),0);
        sort(1+all(sa), [&](int a, int b) { return S[a] < S[b]; });
        F0R(i,1,N) { int a = sa[i-1], b = sa[i];
            isa[b] = i > 1 && S[a] == S[b] ? isa[a] : i; }
        for (int len = 1; len < N; len *= 2) { // currently sorted
            // by first len chars
```

```
        vi s(sa), is(isa), pos(N); iota(all(pos),0);
        each(t,s) {int T=t-len;if (T>=0) sa[pos[isa[T]]++] = T;}
        F0R(i,1,N) { int a = sa[i-1], b = sa[i];
            isa[b] = is[a]==is[b]&&is[a+len]==is[b+len]?isa[a]:i; }
        }
    }
    void genLcp() { // Kasai's Algo
        lcp = vi(N-1); int h = 0;
        F0R(b,N-1) { int a = sa[isa[b]-1];
            while (a+h < sz(S) && S[a+h] == S[b+h]) ++h;
            lcp[isa[b]-1] = h; if (h) h--; }
        R.init(lcp);
    }
    RMQ<int> R;
    int getLCP(int a, int b) { // lcp of suffixes starting at a,b
        if (a == b) return sz(S)-a;
        int l = isa[a], r = isa[b]; if (l > r) swap(l,r);
        return R.query(l,r-1);
    }
};
```

SuffixArrayLinear.h

**Description:** Linear-time suffix array.  
**Usage:** sa.is(s, 26) // all entries must be in [0, 26)  
**Time:**  $\mathcal{O}(N)$ , ~100ms for  $N = 5 \cdot 10^5$

ed0bb4, 46 lines

```
vi sa_is(const vi& s, int upper) {
    int n = sz(s); if (!n) return {};
    vi sa(n); vb ls(n);
    R0F(i,n-1) ls[i] = s[i] == s[i+1] ? ls[i+1] : s[i] < s[i+1];
    vi sum_l(upper), sum_s(upper);
    F0R(i,n) (ls[i] ? sum_l[s[i]+1] : sum_s[s[i]]++)++;
    F0R(i,upper) {
        if (i) sum_l[i] += sum_s[i-1];
        sum_s[i] += sum_l[i];
    }
    auto induce = [&](const vi& lms) {
        fill(all(sa),-1);
        vi buf = sum_s;
        for (int d: lms) if (d != n) sa[buf[s[d]]++] = d;
        buf = sum_l; sa[buf[s[n-1]]++] = n-1;
        F0R(i,n) {
            int v = sa[i]-1;
            if (v >= 0 && !ls[v]) sa[buf[s[v]]++] = v;
        }
        buf = sum_l;
        R0F(i,n) {
            int v = sa[i]-1;
            if (v >= 0 && ls[v]) sa[--buf[s[v]+1]] = v;
        }
    };
    vi lms_map(n+1,-1), lms; int m = 0;
    F0R(i,1,n) if (!ls[i-1] && ls[i]) lms_map[i]=m++, lms.pb(i);
    induce(lms); // sorts LMS prefixes
    vi sorted_lms;each(v,sa)if (lms_map[v]!=-1)sorted_lms.pb(v);
    vi rec_s(m); int rec_upper = 0; // smaller subproblem
    F0R(i,1,m) { // compare two lms substrings in sorted order
        int l = sorted_lms[i-1], r = sorted_lms[i];
        int end_l = lms_map[l]+1 < m ? lms[lms_map[l]+1] : n;
        int end_r = lms_map[r]+1 < m ? lms[lms_map[r]+1] : n;
        bool same = 0; // whether lms substrings are same
        if (end_l-1 == end_r-r) {
            for (;l < end_l && s[l] == s[r]; ++l,++r);
            if (l != n && s[l] == s[r]) same = 1;
        }
        rec_s[lms_map[sorted_lms[i]]] = (rec_upper += !same);
    }
    vi rec_sa = sa_is(rec_s,rec_upper+1);
    F0R(i,m) sorted_lms[i] = lms[rec_sa[i]];
```

```
    induce(sorted_lms); // sorts LMS suffixes
    return sa;
}
```

TandemRepeats.h

**Description:** Find all  $(i,p)$  such that  $s.substr(i,p) == s.substr(i+p,p)$ . No two intervals with the same period intersect or touch.  
**Usage:** solve("aaabababa") // {{0, 1, 1}, {2, 5, 2}}  
**Time:**  $\mathcal{O}(N \log N)$

"SuffixArray.h" 661326, 13 lines

```
V<AR<int,3>> solve(str s) {
    int N = sz(s); SuffixArray A,B;
    A.init(s); reverse(all(s)); B.init(s);
    V<AR<int,3>> runs;
    for (int p = 1; 2*p <= N; ++p) { // do in  $\mathcal{O}(N/p)$  for period p
        for (int i = 0, lst = -1; i+p <= N; i += p) {
            int l = i-B.getLCP(N-i-p,N-i), r = i-p+A.getLCP(i,i+p);
            if (l > r || l == lst) continue;
            runs.pb({lst = l,r,p}); // for each i in [l,r],
        } // s.substr(i,p) == s.substr(i+p,p)
    }
    return runs;
}
```

9.2 Heavy

PalTree.h

**Description:** Used infrequently. Palindromic tree computes number of occurrences of each palindrome within string. ans[i][0] stores min even  $x$  such that the prefix  $s[1..i]$  can be split into exactly  $x$  palindromes, ans[i][1] does the same for odd  $x$ .  
**Time:**  $\mathcal{O}(N \sum)$  for addChar,  $\mathcal{O}(N \log N)$  for updAns

8a7d31, 41 lines

```
struct PalTree {
    static const int ASZ = 26;
    struct node {
        AR<int,ASZ> to = AR<int,ASZ>();
        int len, link, oc = 0; // # occurrences of pal
        int slink = 0, diff = 0;
        AR<int,2> seriesAns;
        node(int _len, int _link) : len(_len), link(_link) {}
    };
    str s = "@"; V<AR<int,2>> ans = {{0,MOD}};
    V<node> d = {{0,1},{-1,0}}; // dummy pals of len 0,-1
    int last = 1;
    int getLink(int v) {
        while (s[sz(s)-d[v].len-2] != s.bk) v = d[v].link;
        return v;
    }
    void updAns() { // serial path has  $\mathcal{O}(\log n)$  vertices
        ans.pb({MOD,MOD});
        for (int v = last; d[v].len > 0; v = d[v].slink) {
            d[v].seriesAns=ans[sz(s)-1-d[v].slink].len-d[v].diff;
            if (d[v].diff == d[d[v].link].diff)
                F0R(i,2) ckmin(d[v].seriesAns[i],
                    d[d[v].link].seriesAns[i]);
            // start of previous oc of link[v]=start of last oc of v
            F0R(i,2) ckmin(ans.bk[i],d[v].seriesAns[i^1]+1);
        }
    }
    void addChar(char C) {
        s += C; int c = C-'a'; last = getLink(last);
        if (!d[last].to[c]) {
            d.eb(d[last].len+2,d[getLink(d[last].link)].to[c]);
            d[last].to[c] = sz(d)-1;
            auto& z = d.bk; z.diff = z.len-d[z.link].len;
            z.slink = z.diff == d[z.link].diff
                ? d[z.link].slink : z.link;
```

```
    } // max suf with different dif
    last = d[last].to[c]; ++d[last].oc;
    updAns();
}
void numOc() { ROF(i,2,sz(d)) d[d[i].link].oc += d[i].oc; }
};
```

SuffixAutomaton.h

**Description:** Used infrequently. Constructs minimal deterministic finite automaton (DFA) that recognizes all suffixes of a string. len corresponds to the maximum length of a string in the equivalence class, pos corresponds to the first ending position of such a string, lnk corresponds to the longest suffix that is in a different class. Suffix links correspond to suffix tree of the reversed string!

**Time:**  $\mathcal{O}(N \log \Sigma)$

a99c6d, 67 lines

```
struct SuffixAutomaton {
    int N = 1; vi lnk{-1}, len{0}, pos{-1}; // suffix link,
    // max length of state, last pos of first occurrence of state
    V<map<char,int>> nex{1}; V<bool> isClone{0};
    // transitions, cloned -> not terminal state
    V<vi> iLnk; // inverse links
    int add(int p, char c) { // ~p nonzero if p != -1
        auto getNext = [&]() {
            if (p == -1) return 0;
            int q = nex[p][c]; if (len[p]+1 == len[q]) return q;
            int clone = N++; lnk.pb(lnk[q]); lnk[q] = clone;
            len.pb(len[p]+1), nex.pb(nex[q]),
            pos.pb(pos[q]), isClone.pb(1);
            for (; ~p && nex[p][c] == q; p = lnk[p]) nex[p][c]=clone;
            return clone;
        };
        // if (nex[p].count(c)) return getNext();
        // ^ need if adding > 1 string
        int cur = N++; // make new state
        lnk.eb(), len.pb(len[p]+1), nex.eb(),
        pos.pb(pos[p]+1), isClone.pb(0);
        for (; ~p && !nex[p].count(c); p = lnk[p]) nex[p][c] = cur;
        int x = getNext(); lnk[cur] = x; return cur;
    }
    void init(str s) { int p = 0; each(x,s) p = add(p,x); }
    // inverse links
    void genIlnk() { iLnk.rsz(N);FOR(v,1,N) iLnk[lnk[v]].pb(v); }
    // APPLICATIONS
    void getAllOccur(vi& oc, int v) {
        if (!isClone[v]) oc.pb(pos[v]); // terminal position
        each(u,iLnk[v]) getAllOccur(oc,u); }
    vi allOccur(str s) { // get all occurrences of s in automaton
        int cur = 0;
        each(x,s) {
            if (!nex[cur].count(x)) return {};
            cur = nex[cur][x]; }
        // convert end pos -> start pos
        vi oc; getAllOccur(oc,cur); each(t,oc) t += 1-sz(s);
        sort(all(oc)); return oc;
    }
    vl distinct;
    ll getDistinct(int x) {
        // # distinct strings starting at state x
        if (distinct[x]) return distinct[x];
        distinct[x]=1;each(y,nex[x]) distinct[x]+=getDistinct(y.s);
        return distinct[x]; }
    ll numDistinct() { // # distinct substrings including empty
        distinct.rsz(N); return getDistinct(0); }
    ll numDistinct2() { // assert(numDistinct()==numDistinct2());
        ll ans = 1; FOR(i,1,N) ans += len[i]-len[lnk[i]];
        return ans; }
};
```

SuffixAutomaton SuffixTree CircularLCS SMAWK

```
SuffixAutomaton S;
vi sa; str s;
void dfs(int x) {
    if (!S.isClone[x]) sa.pb(sz(s)-1-S.pos[x]);
    V<pair<char,int>> chr;
    each(t,S.iLnk[x]) chr.pb({s[S.pos[t]-S.len[x]],t});
    sort(all(chr)); each(t,chr) dfs(t.s);
}
```

```
int main() {
    re(s); reverse(all(s));
    S.init(s); S.genIlnk();
    dfs(0); ps(sa); // generating suffix array for s
}
```

SuffixTree.h

**Description:** Used infrequently. Ukkonen’s algorithm for suffix tree. Longest non-unique suffix of s has length len[p]+lef after each call to add terminates. Each iteration of loop within add decreases this quantity by one.

**Time:**  $\mathcal{O}(N \log \Sigma)$

39751c, 51 lines

```
struct SuffixTree {
    str s; int N = 0;
    vi pos, len, lnk; V<map<char,int>> to;
    int make(int POS, int LEN) { // lnk[x] is meaningful when
        // x!=0 and len[x] != MOD
        pos.pb(POS);len.pb(LEN);lnk.pb(-1);to.eb();return N++; }
    void add(int& p, int& lef, char c) { // longest
        // non-unique suffix is at node p with lef extra chars
        s += c; ++lef; int lst = 0;
        for (;lef;p=lnk[p]:lef--) { // if p != root then lnk[p]
            // must be defined
            while (lef>1 && lef>len[to[p][s[sz(s)-lef]]])
                p = to[p][s[sz(s)-lef]], lef -= len[p];
            // traverse edges of suffix tree while you can
            char e = s[sz(s)-lef]; int& q = to[p][e];
            // next edge of suffix tree
            if (!q) q = make(sz(s)-lef,MOD), lnk[lst] = p, lst = 0;
            // make new edge
            else {
                char t = s[pos[q]+lef-1];
                if (t == c) { lnk[lst] = p; return; } // suffix not
                ↪unique
                int u = make(pos[q],lef-1);
                // new node for current suffix-1, define its link
                to[u][c] = make(sz(s)-1,MOD); to[u][t] = q;
                // new, old nodes
                pos[q] += lef-1; if (len[q] != MOD) len[q] -= lef-1;
                q = u, lnk[lst] = u, lst = u;
            }
        }
    }
    void init(str _s) {
        make(-1,0); int p = 0, lef = 0;
        each(c,_s) add(p,lef,c);
        add(p,lef,'$'); s.pop_back(); // terminal char
    }
    int maxPre(str x) { // max prefix of x which is substring
        for (int p = 0, ind = 0;;) {
            if (ind == sz(x) || !to[p].count(x[ind])) return ind;
            p = to[p][x[ind]];
            FOR(i,len[p]) {
                if (ind == sz(x) || x[ind] != s[pos[p]+i]) return ind;
                ind ++;
            }
        }
    }
    vi sa; // generate suffix array
    void genSa(int x = 0, int Len = 0) {
```

```
        if (!sz(to[x])) sa.pb(pos[x]-Len); // found terminal node
        else each(t,to[x]) genSa(t.s,Len+len[x]);
    }
};
```

Various (10)

10.1 Dynamic programming

When doing DP on intervals:  
 $a[i][j] = \min_{i < k < j} (a[i][k] + a[k][j]) + f(i, j)$ , where the (minimal) optimal  $k$  increases with both  $i$  and  $j$ ,

- one can solve intervals in increasing order of length, and search  $k = p[i][j]$  for  $a[i][j]$  only between  $p[i][j - 1]$  and  $p[i + 1][j]$ .
- This is known as Knuth DP. Sufficient criteria for this are if  $f(b, c) \leq f(a, d)$  and  $f(a, c) + f(b, d) \leq f(a, d) + f(b, c)$  for all  $a \leq b \leq c \leq d$ .
- Consider also: LineContainer (ch. Data structures), monotone queues, ternary search.

CircularLCS.h

**Description:** Used only twice. For strs  $A, B$  calculates longest common subsequence of  $A$  with all rotations of  $B$

**Time:**  $\mathcal{O}(|A| \cdot |B|)$

db21cf, 26 lines

```
int circular_lcs(str A, str B) {
    B += B;
    int max_lcs = 0;
    V<vb> dif_left(sz(A)+1,vb(sz(B)+1)), dif_up(sz(A)+1,vb(sz(B)
    ↪+1));
    auto recalc = [&](int x, int y) { assert(x && y);
        int res = (A.at(x-1) == B.at(y-1)) |
            dif_up[x][y-1] | dif_left[x-1][y];
        dif_left[x][y] = res-dif_up[x][y-1];
        dif_up[x][y] = res-dif_left[x-1][y];
    };
    FOR(i,1,sz(A)+1) FOR(j,1,sz(B)+1) recalc(i, j);
    FOR(j,sz(B)/2) {
        // 1. zero out dp[.][j], update dif_left and dif_right
        if (j) for (int x = 1, y = j; x <= sz(A) && y <= sz(B); ) {
            int pre_up = dif_up[x][y];
            if (y == j) dif_up[x][y] = 0;
            else recalc(x,y);
            (pre_up == dif_up[x][y]) ? ++x : ++y;
        }
        // 2. calculate LCS(A[0:sz(A)),B[j:j+sz(B)/2])
        int cur_lcs = 0;
        FOR(x,1,sz(A)+1) cur_lcs += dif_up[x][j+sz(B)/2];
        ckmx(max_lcs,cur_lcs);
    }
    return max_lcs;
}
```

SMAWK.h

**Description:** Given negation of totally monotone matrix with entries of type D, find indices of row maxima (their indices increase for every submatrix). If tie, take lesser index. f returns matrix entry at  $(r, c)$  in  $\mathcal{O}(1)$ . Use in place of divide & conquer to remove a log factor.

**Time:**  $\mathcal{O}(R + C)$ , can be reduced to  $\mathcal{O}(C(1 + \log R/C))$  evaluations of  $f$

0a9abb, 25 lines

```
template<class F, class D=ll> vi smawk (F f, vi x, vi y) {
    vi ans(sz(x),-1); // x = rows, y = cols
```

```
#define upd() if (ans[i] == -1 || w > mx) ans[i] = c, mx = w
if (min(sz(x),sz(y)) <= 8) {
    F0R(i,sz(x)) { int r = x[i]; D mx;
        each(c,y) { D w = f(r,c); upd(); } }
    return ans;
}
if (sz(x) < sz(y)) { // reduce subset of cols to consider
    vi Y; each(c,y) {
        for (;sz(Y);Y.pop_back()) { int X = x[sz(Y)-1];
            if (f(X,Y.bk) >= f(X,c)) break; }
        if (sz(Y) < sz(x)) Y.pb(c);
    } y = Y;
} // recurse on half the rows
vi X; for (int i = 1; i < sz(x); i += 2) X.pb(x[i]);
vi ANS = smawk(f,X,y); F0R(i,sz(ANS)) ans[2*i+1] = ANS[i];
for (int i = 0, k = 0; i < sz(x); i += 2){
    int to = i+1 < sz(ans) ? ans[i+1] : y.bk; D mx;
    for(int r = x[i];++k) {
        int c = y[k]; D w = f(r,c); upd();
        if (c == to) break; }
}
return ans;
};
```

10.2 Debugging tricks

- signal(SIGSEGV, [](int) { \_Exit(0); }); converts segfaults into Wrong Answers. Similarly one can catch SIGABRT (assertion failures) and SIGFPE (zero divisions). \_GLIBCXX\_DEBUG violations generate SIGABRT (or SIGSEGV on gcc 5.4.0 apparently).
- feenableexcept (29); kills the program on NaNs (1), 0-divs (4), infinities (8) and denormals (16).

10.3 Optimization tricks

10.3.1 Bit hacks

- x & -x is the least bit in x.
- for (int x = m; x; ) { --x &= m; ... } loops over all subset masks of m (except m itself).
- c = x&-x, r = x+c; ((r^x) >> 2)/c) | r is the next number after x with the same number of bits set.
- F0R(b,k) F0R(i,1<<K) if (i&1<<b) D[i] += D[i^(1<<b)]; computes all sums of subsets.

10.3.2 Pragmas

- #pragma GCC optimize ("Ofast") will make GCC auto-vectorize for loops and optimizes floating points better (assumes associativity and turns off denormals).
- #pragma GCC target ("avx,avx2") can double performance of vectorized code, but causes crashes on old machines. Also consider older #pragma GCC target ("sse4").
- #pragma GCC optimize ("trapv") kills the program on integer overflows (but is really slow).

FastIO.h

Description: Fast input and output for integers and strings. For doubles, read them as strings and convert them to double using stod.

Usage: initO(); int a,b; ri(a,b); wi(b,'n'); wi(a,'\n');

Time: input is ~300ms faster for 10<sup>6</sup> long longs on CF.

cbf279, 39 lines

```
inline namespace FastIO {
const int BSZ = 1<<15; // INPUT
char ibuf[BSZ]; int ipos, ilen;
char nc() { // next char
    if (ipos == ilen) {
        ipos = 0; ilen = fread(ibuf,1,BSZ,stdin);
        if (!ilen) return EOF;
    }
    return ibuf[ipos++];
}
void rs(str& x) { // read str
    char ch; while (isspace(ch = nc()));
    do { x += ch; } while (!isspace(ch = nc()) && ch != EOF);
}
tcT> void ri(T& x) { // read int or ll
    char ch; int sgn = 1;
    while (!isdigit(ch = nc())) if (ch == '-') sgn *= -1;
    x = ch-'0'; while (isdigit(ch = nc())) x = x*10+(ch-'0');
    x *= sgn;
}
tcT, class... Ts> void ri(T& t, Ts&... ts) {
    ri(t); ri(ts...); } // read ints
///// OUTPUT (call initO() at start)
char obuf[BSZ], numBuf[100]; int opos;
void flushOut() { fwrite(obuf,1,opos,stdout); opos = 0; }
void wc(char c) { // write char
    if (opos == BSZ) flushOut();
    obuf[opos++] = c; }
void ws(str s) { each(c,s) wc(c); } // write str
tcT> void wi(T x, char after = '\0') {
    if (x < 0) wc('-'), x *= -1;
    int len = 0; for (;x>=10;x/=10) numBuf[len++] = '0'+(x%10);
    wc('0'+x); R0F(i,len) wc(numBuf[i]);
    if (after) wc(after);
}
void initO() { assert(atexit(flushOut) == 0); }

}
```

10.4 Other languages

Python3.py

Description: Python review.

40 lines

```
from math import *
import sys, random
def nextInt():
    return int(input())
def nextStrs():
    return input().split()
def nextInts():
    return list(map(int,nextStrs()))

n = nextInt()
v = [n]
def process(x):
    global v
    x = abs(x)
    V = []
    for t in v:
        g = gcd(t,x)
        if g != 1:
            V.append(g)
```

```
if g != t:
    V.append(t//g)
v = V
for i in range(50):
    x = random.randint(0,n-1)
    if gcd(x,n) != 1:
        process(x)
    else:
        sx = x*x%n # assert(gcd(sx,n) == 1)
        print(f"sqrt {sx}")
        sys.stdout.flush()
        X = nextInt()
        process(x+X)
        process(x-X)
print(f'! {len(v)}',end='')
for i in v:
    print(f' {i}',end='')
print()
sys.stdout.flush() # sys.exit(0) -> exit
# sys.setrecursionlimit(int(1e9)) -> stack size
# print(f'{ans:.6f}') -> print ans to 6 decimal places
```

Decimal.py

Description: Arbitrary-precision decimals

5 lines

```
from decimal import *

getcontext().prec = 100 # how many digits of precision
print(Decimal(1) / Decimal(7)) # 0.142857142857...
print(Decimal(10) ** -100) # 1E-100
```